

Odd-Cycle-Free Latin Squares: Products, Congruence Obstructions, and Constructed Families

28 September 2026

Reviewed research update; compact companion

Not externally peer reviewed

Abstract

A Latin square is FFF when every two-line permutation in each of its row, column and symbol views has only even cycles. Direct products take the componentwise Boolean OR of the three failure patterns and are group-isotopic exactly when both factors are. Subsquare and congruence obstructions restrict possible constructions. Exact small-order checks exclude orders 6 and 10, while independently checked tables of orders 12 and 14 refute the former power-of-two existence conjecture. Finite-field constructions give additional infinite even-order families. An affine prime-successor family has an exact three-view criterion and parameter classification; a mixed-character-sum estimate gives more than $p/6000$ represented FFF classes at order $p + 1$ for every prime $p \geq 10^{12}$. This does not determine the complete order spectrum.

A row-dependent extension has pattern equal to the OR of its quotient and fibre patterns. When its fibres have no binary three-sorted quotient, all binary quotients descend from the outer square. For an elementary abelian 2-group as outer square, this recovers the layers intrinsically and classifies a fixed family of fibre assignments by affine orbits. A finite audit supplies 1703 pairwise distinct FFF main classes of order 20; the construction and Burnside's lemma then give explicit lower bounds in orders $20 \cdot 2^d$, not censuses of those orders. Direct binary-rank checks establish quotient freeness for these fixed order-20 inputs, independently of the computational order-10 exclusion.

For cell-dependent block substitutions, an exact return-map formula explains how an odd-cycle quotient can nevertheless lift to an FFF table. A common binary-field fibre constructs FFF squares of order $N2^D$ for every odd cubefree N when $\text{ord}_{\text{rad}(N)}(2) \mid D$ and $2^D > 3(N - 1)$. A distinguished affine corner trade preserves all three pattern bits and produces FFF tables with exactly one intercalate, including explicit orders 14 and 20. These constructions and their necessary boundaries do not decide order 18.

An exact position-value coloring formulation supplies a smaller three-view constraint representation with automatic color balance. Proofs, finite computations, and bounded search outcomes are distinguished throughout. Existence at order 18, the complete spectrum, and literature priority for the recent constructions remain unresolved.

Keywords. Latin square; quasigroup; Latin trade; direct product; main class; affine orbit; computational enumeration.

2020 Mathematics Subject Classification. Primary 05B15; Secondary 20N05, 05C70.

Status, attribution, and evidence boundary

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The evidence intake for this edition is 28 September 2026 at 10:24 UTC, through register entry C271. It incorporates the preceding review corrections and selected completed results after C251. Work completed after this fixed intake is not silently incorporated. Earlier source packages, claims and numerical records are preserved. This is a new publicly distributable research snapshot, not journal acceptance.

Theorems with written proofs are distinguished from statements labelled *Computational Theorem*, which require exact finite evidence in addition to a mathematical reduction. Explicitly scoped local exclusions and timeouts are not global theorems. Internal AI-assisted proof review, independent implementations, and machine proof checking are three different forms of checking; none is independent human refereeing. The compact companion imports the order-10 exclusion proved in the full report as an explicit computational dependency. Neither document resolves order 18 or asserts a complete higher-order main-class census.

1 Introduction

An odd cycle in the permutation induced by two lines of a Latin square supports a simple trade: swap the two lines on that cycle. The move preserves the Latin property. In the row and column views it reverses the conventional row–column Latin sign; in the symbol view it preserves that sign and reverses the two parastrophically transported signs. Wanless already gives the underlying three line-parity changes for cycle switches [Wan04, Proposition 1]; the pairwise sign products here are their explicit corollary. This local operation is therefore a natural primitive in parity-based approaches related to the Alon–Tarsi setting, but its three sign conventions must be kept distinct. Bérczi’s experimental program studies such local corrections as ingredients for global constructions [Bér26]; it does not supply the theorems proved here.

We isolate the underlying existence question. For each of the row, column, and symbol views, ask whether some pair of lines induces a permutation with a nontrivial odd cycle. The three answers form a pattern in $\{F, T\}^3$. A square is FFF when all three answers are negative. Equivalently, every union of two link factors in the tripartite representation has component lengths divisible by four. The two-factor/cycle correspondence in one view is classical [WI05, Section 3]; the three-view formulation follows by conjugating the coordinate roles.

The structural results start with two elementary boundaries. Every Latin square of odd order greater than one is TTT, whereas the Cayley table of a finite group is FFF exactly when the group is a 2-group. A basis-family closure criterion detects group isotopy. We then show that failure patterns are monotone from Latin subsquares and analyze quasigroup congruences: odd block size forces TTT, while two-element blocks preserve the quotient pattern exactly. Thus a hypothetical FFF quasigroup of order $2p$, for odd prime p , must be congruence-simple in every isotope. This excludes all congruence-imprimitive

constructions at order 10 but does not by itself decide the simple case.

The remaining binary line-incidence defect also has an exact subsquare meaning. Nonuniversal vectors in the left binary line kernel are in bijection with half-order subsquares, and the quotient classes group them into complementary four-packs. This yields a closed counting formula and forces maximal binary line rank for every FFF order congruent to 2 modulo 4 and greater than 2. At order 10 this is necessary, not sufficient.

More generally, Moorhouse’s 3-net character formula specializes as follows: a nonuniversal line dependency over a prime field is a balanced cyclic quotient homotopy, and after isotopy its fibres form a quasigroup congruence. Consequently every Smith torsion prime divides the order. For order $2p$, with p odd prime, the congruence obstruction shows that even one F view forces a saturated integer line-incidence lattice.

The group construction is not the whole phenomenon. A twisted binary extension of C_4 gives an explicit non-group-isotopic FFF loop of order 8. Moreover, direct products satisfy the exact componentwise formula

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K),$$

and a product is group-isotopic exactly when both factors are. These theorems turn one order-8 example into infinite power-of-two families.

The order-8 computation scans the cited main-class census. McKay, Meynert, and Myrvold enumerated the main classes of small Latin squares [MMM07]; the associated ANU file contains one representative of each of the 283657 order-8 main classes [McK]. The scan finds exactly 230 FFF main classes. Only 5 are group-isotopic and 225 are not, and all eight ordered failure patterns occur. Independently, fresh exact-cover generation of all 9408 reduced order-6 squares finds no FFF square and matches the official ANU set exactly.

Combining theory and computation proves that every order 2^m , $m \geq 3$, realizes all eight patterns and contains a non-group-isotopic FFF square. We do not claim unrestricted direct-product cancellation on isotopy or main classes. Section 10 establishes more specific distinction and classification results under explicit hypotheses. The accompanying complete research report proves by exhaustive master searches that order 10 has no FFF square; its proof is not repeated in this compact companion, but its result is explicitly used in $h(5) = 2$. The fixed order-20 family instead has direct binary-rank certificates. A reduced order-12 FFF witness, independently checked in all three views, refutes the former power-of-two conjecture; orders 14 and 36 have further positive witnesses. The classical prime round-robin table is FFF exactly for $p \equiv 3 \pmod{8}$, supplying infinitely many non-power-of-two orders; with the separately audited order-10 exclusion, its order-20 instance gives $h(5) = 2$. A written three-point lift proof gives an FFF square of order $m2^{\text{ord}_m(2)}$ for every odd $m > 1$, without determining the least dyadic exponent in general. Order 18 remains the next undecided small even non-power-of-two order.

The row-fibred extension in Section 10 generalizes the product pattern formula. Quotient-free fibres make the quotient layers intrinsic; over an elementary abelian outer group this gives an exact affine-orbit classification within a fixed family. Exact finite controls supply first 514, and in the present enlarged audit 1703, order-20 input classes. The resulting large counts come from a construction theorem and Burnside’s lemma, not direct scans of every output square.

The longer companion develops additional cycle-parity, subset-action, primitive-group, and flag-surface machinery, gives the order-10 completeness argument, and records the restricted order-18 search boundary. The general product, row-fibred, and quotient-descent proofs and their stated fixed-family order-20 specialization do not depend on that

machinery. The claims of order-10 nonexistence and $h(5) = 2$ retain the complete order-10 computational dependency.

The present update adds the general block-return cycle formula, a common binary-field construction over every odd cubefree quotient, and an affine corner trade preserving the three pattern bits. The last gives FFF tables with a unique intercalate and strengthens the order-40 class lower bound to 1450956. All these are separate from the unresolved order-18 search.

2 Definitions and invariance

Let X be a set of size n . A *Latin square* on X is a map $L: X \times X \rightarrow X$ such that every row and every column is a bijection. For $a, c, u \in X$, define the row, column, and symbol-line bijections

$$r_a(c) = L(a, c), \quad q_c(a) = L(a, c), \quad s_u(c) = \text{the unique } a \text{ such that } L(a, c) = u.$$

For two lines in the corresponding view, put

$$\rho_{a,b} = r_b^{-1} r_a, \quad \kappa_{c,d} = q_d^{-1} q_c, \quad \sigma_{u,v} = s_v^{-1} s_u.$$

All three permutations act on an n -element complementary coordinate set.

Lemma 2.1. *If the two lines are distinct, the corresponding induced permutation is fixed-point-free.*

Proof. A fixed point of $\rho_{a,b}$ would give $L(a, c) = L(b, c)$ in one column, contrary to the Latin property. The column case is identical. A fixed point of $\sigma_{u,v}$ would say that the same cell contains the two distinct symbols u and v . $\square$

Definition 2.2. A view of L *fails* if one of its induced two-line permutations has an odd cycle of length greater than one. Write

$$\text{pat}(L) = (\text{row}(L), \text{col}(L), \text{sym}(L)) \in \{F, T\}^3,$$

where T means that the view fails. A square is *FFF* if $\text{pat}(L) = \text{FFF}$.

The terminology “fails” is chosen from the perspective of the local-trade program: a T view supplies an odd-cycle trade, while an F view does not.

Definition 2.3. Fix two rows $a \neq b$ and a union C of cycles of $\rho_{a,b}$. The associated *two-line trade* swaps the entries in rows a, b in precisely the columns in C . Column- and symbol-view trades are defined by permuting the coordinate roles.

Proposition 2.4. *A two-line trade produces another Latin square. Define the three line-sign products*

$$R(L) = \prod_a \text{sgn}(r_a), \quad C(L) = \prod_c \text{sgn}(q_c), \quad S(L) = \prod_u \text{sgn}(s_u),$$

and put

$$\epsilon_{RC} = RC, \quad \epsilon_{RS} = RS, \quad \epsilon_{CS} = CS.$$

The first is the conventional row-column Latin sign, while the other two are its parastrophic companions. If a trade has support size k and $\delta = (-1)^k$, then its sign ratios are

trade view	R'/R	C'/C	S'/S	$\epsilon'_{RC}/\epsilon_{RC}$	$\epsilon'_{RS}/\epsilon_{RS}$	$\epsilon'_{CS}/\epsilon_{CS}$
row	1	δ	δ	δ	δ	1
col	δ	1	δ	δ	1	δ
sym	δ	δ	1	1	δ	δ

Consequently an odd row- or column-view trade reverses the conventional ϵ_{RC} , whereas an odd symbol-view trade preserves ϵ_{RC} and reverses ϵ_{RS} and ϵ_{CS} .

Proof. Write $\pi = \rho_{a,b}$. Since $r_a(c) = r_b(\pi(c))$ and C is π -invariant, the two rows contain the same symbol set on C . Swapping those entries therefore preserves both row symbol sets, while each affected column merely swaps two entries. Thus the result is Latin.

Extend $\pi|_C$ by the identity outside C . The two changed row permutations are obtained by precomposition with π and π^{-1} , so their sign changes cancel and $R'/R = 1$. Each of the $|C|$ affected column permutations is composed with a transposition, so $C'/C = (-1)^{|C|}$. The same support contains the same $|C|$ symbols in the two rows; each associated symbol-position permutation swaps those two rows, giving $S'/S = (-1)^{|C|}$. This proves the first table row. Exchanging rows and columns proves the second.

For a symbol-view trade, each support column swaps the two selected symbols, and the cycle relation pairs the affected cells in exactly the same number of rows. Hence $R'/R = C'/C = (-1)^{|C|}$. The two changed symbol-position permutations differ from their predecessors by mutually inverse permutations on the support, so their sign changes cancel and $S'/S = 1$. Multiplying the appropriate line-sign ratios proves the last three columns. $\square$

Proposition 2.5 (Tripartite link formulation). *Regard the cells of L as the triangle decomposition*

$$\{\{a, c, L(a, c)\} : a, c \in X\}$$

of the complete tripartite graph on the row, column, and symbol copies of X . The links of the vertices in each part form a 1-factorization between the other two parts. An induced two-line cycle of length ℓ corresponds bijectively to an alternating cycle of length 2ℓ in the union of the two associated link factors. Hence L is FFF if and only if every two-factor union in all three link factorizations has only cycle lengths divisible by four.

Furthermore, the total number of 4-cycles among the two-factor unions is the same in all three views and equals the number of intercalates of L .

Proof. For a row a , its incident triangles give the perfect matching $M_a = \{\{c, r_a(c)\} : c \in X\}$ between columns and symbols. For distinct rows a, b , alternately following an edge of M_a and an edge of M_b sends a column c to $r_b^{-1}r_a(c) = \rho_{a,b}(c)$. Thus an ℓ -cycle of $\rho_{a,b}$ is exactly one alternating 2ℓ -cycle of $M_a \cup M_b$. Cyclically permuting the three coordinate classes proves the column and symbol statements and the FFF equivalence.

A 4-cycle in $M_a \cup M_b$ is exactly a 2-cycle of $\rho_{a,b}$, hence a 2-by-2 Latin subsquare on two rows, two columns, and two symbols. This is an intercalate. The same intercalate is counted once by its row pair, once by its column pair, and once by its symbol pair, proving the three equalities. $\square$

Lemma 2.6 (Global three-view sign identity). *For every Latin square of order n ,*

$$R(L)C(L)S(L) = (-1)^{\binom{n}{2}}.$$

Proof. Give each coordinate copy of X the same fixed order, and let $\tau(x, y) = (y, x)$ on X^2 . The $\binom{n}{2}$ off-diagonal pairs are the transpositions of τ , so $\text{sgn}(\tau) = (-1)^{\binom{n}{2}}$.

The row-entry map $A(r, c) = (r, L(r, c))$ is block diagonal by rows and has sign $R(L)$. The map $B(r, c) = (c, L(r, c))$ is a coordinate transposition followed by the block-diagonal family of column permutations, so its sign is $(-1)^{\binom{n}{2}}C(L)$. Likewise, $D(c, s) = (s, s_s(c))$ has sign $(-1)^{\binom{n}{2}}S(L)$. Since $A = \tau \circ D \circ B$, taking signs gives

$$R(L) = (-1)^{3\binom{n}{2}}S(L)C(L) = (-1)^{\binom{n}{2}}S(L)C(L),$$

which is equivalent to the claimed identity. $\square$

For later comparison, let $C_{v,k}$ denote the total number of k -cycles among all induced line-pair permutations in view v .

Proposition 2.7 (Aggregate cycle-profile baseline). *Let n be even and let p be prime. If*

$$\sum_{k=2}^n a_k x_k + b = 0 \pmod{p}$$

for every cycle-count vector $(x_2, \dots, x_n)$ of a fixed-point-free permutation of degree n , then, separately for each view v ,

$$\sum_{k=2}^n a_k C_{v,k} + \binom{n}{2} b = 0 \pmod{p}.$$

Moreover,

$$C_{\text{row},2} = C_{\text{col},2} = C_{\text{sym},2}$$

and

$$\sum_{v \in \{\text{row}, \text{col}, \text{sym}\}} \sum_{\substack{2 \leq k \leq n \\ k \text{ even}}} C_{v,k} = \frac{n(n-1)}{2} \pmod{2}.$$

Proof. The aggregate profile in one view is the sum of the $\binom{n}{2}$ fixed-point-free line-pair profiles, proving the first identity. The second identity is the intercalate bijection from Proposition 2.5. For the last identity, the sign of a fixed-point-free permutation of even degree is $(-1)^e$, where e is its number of even cycles. Multiplying relative-permutation signs over all line pairs in a view gives the product of its line signs. Lemma 2.6 gives the product over the three views as $(-1)^{n(n-1)/2}$, which proves the congruence. $\square$

Proposition 2.8 (Line-pair sign cut). *Fix one view and write its line bijections as λ_i . For $P_{ij} = \lambda_j^{-1} \lambda_i$, the pairs for which P_{ij} is odd form a cut of the complete graph on the lines. If n is even and the view is F , then the pairs for which P_{ij} has an odd number of cycles form the same cut. Consequently, for even $n > 2$, no view can have every induced line-pair permutation equal to one n -cycle.*

Proof. Put $\varepsilon_i = \text{sgn}(\lambda_i)$. Since sign is a homomorphism,

$$\text{sgn}(P_{ij}) = \varepsilon_i \varepsilon_j.$$

Thus P_{ij} is odd exactly when the endpoint signs differ. These pairs are the edges of the cut between positive and negative lines.

In an F view of even degree, every cycle of P_{ij} has even length and therefore negative sign. Hence $\text{sgn}(P_{ij}) = (-1)^{c(P_{ij})}$, where $c(P_{ij})$ is the number of cycles. Finally, if all relative permutations were n -cycles, every edge would be negative. A triangle of three lines would then cross the sign cut three times, whereas every cut meets a cycle evenly. $\square$

An *isotopy* independently permutes rows, columns, and symbols. A *parastrophy* permutes the three coordinate roles. Their combined orbits are the *main classes* (also called species).

Proposition 2.9. *Under isotopy, each of the three view bits is invariant. Under parastrophy, the three bits are permuted. Consequently, being FFF and having at least one failing view are main-class invariants.*

Proof. Let an isotopy act by permutations α, β, γ of rows, columns, and symbols. The transformed row maps satisfy $r'_{\alpha(a)} = \gamma r_a \beta^{-1}$, and hence

$$(r'_{\alpha(b)})^{-1} r'_{\alpha(a)} = \beta(r_b^{-1} r_a) \beta^{-1}.$$

Thus row cycle lengths are unchanged. The column calculation conjugates by α , and the symbol calculation conjugates by β . Parastrophy only permutes which coordinate pair is called row, column, or symbol. $\square$

3 Elementary structural results

3.1 Odd orders and group tables

Proposition 3.1. *Every Latin square of odd order $n > 1$ has pattern TTT.*

Proof. By Lemma 2.1, every induced permutation between two distinct lines is fixed-point-free on an odd set. If all of its nontrivial cycles were even, their lengths would have even sum. Hence it has an odd cycle of length greater than one. This argument applies separately to any pair of distinct rows, columns, and symbols. $\square$

Let G be a finite group and let $L_G(x, y) = xy$ be its Cayley table.

Proposition 3.2. *The row permutation induced by rows a, b of L_G is left multiplication by $b^{-1}a$. The column permutation induced by columns c, d is right multiplication by cd^{-1} . The symbol permutation induced by symbols u, v is right multiplication by $u^{-1}v$. Each cycle length is therefore the order of the corresponding group element.*

Proof. The row identity is $r_b^{-1} r_a(x) = b^{-1}ax$. Likewise $q_d^{-1} q_c(x) = xcd^{-1}$. Finally, $s_u(c) = uc^{-1}$ and $s_v^{-1}(a) = a^{-1}v$, whence $s_v^{-1} s_u(c) = cu^{-1}v$. Left or right multiplication by g has orbits equal to cosets of $\langle g \rangle$, all of length $\text{ord}(g)$. $\square$

Corollary 3.3. *The Cayley table of a finite group G is FFF if and only if G is a 2-group.*

Proof. If G is a 2-group, every nonidentity element has even order, so Proposition 3.2 gives no nontrivial odd cycle. Conversely, if $|G|$ has an odd prime divisor, Cauchy's theorem supplies a nonidentity element of odd order; the corresponding induced permutations give a failing view. $\square$

3.2 A closure criterion for group isotopy

Fix a base row b of a Latin square L and define

$$H_L(b) = \{r_b^{-1} r_x : x \in X\} \subseteq \text{Sym}(X).$$

Proposition 3.4 (Basis-family closure criterion). *A Latin square L is isotopic to a group table if and only if $H_L(b)$ is closed under composition. This condition is independent of the chosen base row.*

Proof. Suppose first that $H = H_L(b)$ is closed. It contains the identity, has $|X|$ distinct elements, and is a finite submonoid of a symmetric group; hence it is a group. For each $z \in X$, the values $(r_b^{-1}r_x)(z)$, as x varies, are all distinct, since a column of L contains every symbol once. Thus H acts regularly on X .

Fix $z_0 \in X$. Relabel rows by $x \mapsto h_x = r_b^{-1}r_x$. Use the regular action to identify every column z with the unique $h_z \in H$ satisfying $h_z(z_0) = z$, and identify every relabeled symbol $r_b^{-1}(u)$ in the same way. The entry in row h_x and column h_z is represented by the unique element sending z_0 to $h_x h_z(z_0)$, namely $h_x h_z$. The resulting isotope is therefore the Cayley table of H .

Conversely, for a group table the family at base row b is the full left regular representation $\{\lambda_{b^{-1}x} : x \in G\}$ and is closed. Isotopy conjugates the family and relabels its members, preserving closure. The same argument for any base row proves base independence. $\square$

The same criterion can equivalently be written in the column or symbol view, by parastrophy.

4 Subsquare and congruence obstructions

The FFF property is hereditary for Latin subsquares in a stronger, componentwise sense.

Theorem 4.1 (Subsquare monotonicity). *If M is a Latin subsquare of L , then*

$$\text{pat}(M) \leq \text{pat}(L)$$

componentwise, with $F < T$.

Proof. Let the rows, columns, and symbols of M be R_0, C_0, S_0 . For $a, b \in R_0$, the ambient row permutation $r_b^{-1}r_a$ preserves C_0 : if $c \in C_0$, then $L(a, c) \in S_0$, and row b contains this symbol at a unique column of C_0 . Its restriction is precisely the induced row permutation of M . Thus every odd row cycle of M is also an odd row cycle of L . Interchanging rows and columns proves the column statement.

For $u, v \in S_0$ and $c \in C_0$, the row $s_u(c)$ lies in R_0 . In that row, the unique occurrence of v lies in a column $d \in C_0$. Hence $s_v^{-1}s_u$ preserves C_0 and restricts to the symbol permutation of M . This proves the symbol statement. $\square$

Corollary 4.2. *Every Latin subsquare of an FFF square is FFF. In particular, an FFF square contains no subsquare of odd order greater than one.*

Proof. The first assertion is Theorem 4.1 with $\text{pat}(L) = \text{FFF}$. The second follows from Proposition 3.1. $\square$

Together with Computational Theorem 6.1, the same implication excludes order-6 subsquares; that additional exclusion inherits the computational status of the order-6 theorem.

There is an exact linear-algebraic description of all half-order subsquares. Let M_L be the $3n$ -by- n^2 binary incidence matrix whose rows are the row, column, and symbol lines and whose columns are the occupied cells of L . Let

$$U = \{(\alpha \mathbf{1}_R, \beta \mathbf{1}_C, \gamma \mathbf{1}_S) : \alpha + \beta + \gamma = 0\} \subseteq \ker M_L^\top$$

and put

$$\delta_2(L) = (3n - 2) - \text{rank}_{\mathbb{F}_2} M_L.$$

Theorem 4.3 (Binary line kernel and half-order subsquares). *Individual embedded half-order Latin subsquares of L are in bijection with*

$$\ker M_L^T \setminus U.$$

Equivalently, the nonzero classes in $\ker M_L^T/U$ parametrize their complementary four-packs. Thus their number is exactly

$$4(2^{\delta_2(L)} - 1). \quad (1)$$

If $n = 2m$, where $m > 1$ is odd, then any one F-view forces

$$\text{rank}_{\mathbb{F}_2} M_L = 3n - 2. \quad (2)$$

In particular, (2) holds for every FFF square of order congruent to 2 modulo 4 and greater than 2.

Proof. A left-kernel vector is a triple (a, b, c) satisfying

$$a_r + b_j + c_{L(r,j)} = 0. \quad (3)$$

If one coordinate, say a , is constant, then fixing a column and varying its rows shows that c is constant; fixing a row then makes b constant. Hence every vector outside U has all three coordinates nonconstant.

Fix a row r . Since that row permutes the symbols, (3) identifies the multiset of c -values with that of $a_r + b$. Choosing rows with $a_r = 0$ and $a_r = 1$ gives

$$\text{wt}(c) = \text{wt}(b) = n - \text{wt}(b).$$

Thus $n = 2m$ and b, c each have weight m . The analogous fixed-column argument gives $\text{wt}(a) = m$. For their binary fibres A_x, B_y, C_z , equation (3) now gives

$$L(A_x, B_y) \subseteq C_{x+y} \quad (x, y \in \mathbb{F}_2).$$

Every set has size m ; Latinness makes each of the four displayed blocks an order- m Latin subsquare.

Adding an element of U complements zero or two of the three partitions and selects a different block in the same four-pack. Conversely, if $R_0 \times C_0$ is a half-order subsquare with symbol set S_0 , row and column counting forces the complementary pattern

$$R_0 \times C_0, R_1 \times C_1 \longrightarrow S_0, \quad R_0 \times C_1, R_1 \times C_0 \longrightarrow S_1.$$

The three complement indicators satisfy (3), and the chosen block recovers them uniquely. This proves both bijections. Since U has dimension two and the quotient has dimension $\delta_2(L)$, counting its nonzero classes proves the counting formula.

Finally suppose that $m > 1$ is odd and $\delta_2(L) > 0$. The preceding construction gives an order- m subsquare. Two distinct rows inside it induce a fixed-point-free permutation preserving its m columns. Such a permutation on an odd set has a nontrivial odd cycle, contradicting an F row view. The two parastrophes give the same conclusion for either other view, which proves the rank formula. The exception $m = 1$ is real: the order-2 FFF square has four singleton subsquares and $\delta_2 = 1$. $\square$

The complete finite audit independently constructs both sides of the bijection on 9778 tables and obtains exact set equality in every case. In particular, the 230 order-8 FFF representatives have

$$\delta_2 = 0, 1, 2, 3 \quad \text{with multiplicities} \quad 63, 156, 10, 1,$$

which explains their half-order-subsquare distribution $0^{63}4^{156}12^{10}28^1$. A hypothetical order-10 FFF square must have binary line rank 28 and no order-5 subsquare. This condition is not sufficient: all 135 tracked order-10 diagnostic tables already satisfy it, none is FFF, and that corpus is not exhaustive.

The same kernel equation has a uniform interpretation over every prime field, now in terms of quotient homotopies rather than individual subsquares. Moorhouse's character formula for loop-coordinatized 3-nets gives the general identity $\text{rank}_p M_L = 3n - 2 - \dim \text{Hom}(G, \mathbb{F}_p)$ and identifies its defect with elementary-abelian quotient characters [Moo91, Theorem 4.2]. We record a direct three-sorted proof in our notation because the individual cyclic quotient is the input needed for the FFF corollary below. The author's publication page flags an erratum for the original article; the reference entry therefore points to the author-hosted article copy and the separate erratum listing. The direct proof given here, rather than an unexamined appeal to the original pagination, is the logical basis for the specialization used below.

Theorem 4.4 (Prime-field line quotients). *Let p be prime and let $(a, b, c) \in \ker_{\mathbb{F}_p} M_L^T$ be nonconstant modulo the two-dimensional constant subspace. Then $p \mid n$, each of a, b, c takes every value exactly n/p times, and their fibres satisfy*

$$L(R_x, C_y) \subseteq S_{-x-y} \quad (x, y \in \mathbb{F}_p).$$

After a suitable isotopy these three fibre partitions become one quasigroup congruence with p blocks of size n/p and quotient isotopic to C_p .

Over characteristic zero, M_L has rank $3n - 2$. Every prime dividing a nonzero Smith invariant factor of the integer matrix M_L divides n .

Proof. If one of a, b, c is constant, fixing a line in either other sort and using Latinness makes the remaining two constant. Let $\mu_b(t)$ and $\mu_c(t)$ be value multiplicities. Since every row permutes the symbols, equation (3), now read in $\mathbb{F}_p$, gives

$$\mu_c(t) = \mu_b(-a_r - t).$$

Comparing rows with $a_r - a_{r'} \neq 0$ makes μ_b invariant under a nonzero translation. That translation generates the additive group of $\mathbb{F}_p$, so μ_b is uniform. Thus $p \mid n$ and both b, c are balanced; the column argument balances a . The displayed fibre rule is the kernel equation itself.

Choose bijections from one partitioned n -set to the row, column, and symbol sets, matching equally labelled fibres. Transporting L by these three bijections gives an isotope whose block operation is $x * y = -x - y$ on $\mathbb{F}_p$. It is Latin and isotopic to addition, so block equivalence is compatible with multiplication and both divisions and is a quasigroup congruence.

In characteristic zero a finite nonempty multiset cannot be invariant under a nonzero translation, because its iterated translates would have infinite support. Hence only the two constant dependencies remain. Finally, if a prime divides a nonzero Smith invariant factor, reduction modulo that prime lowers the rank; the first part then forces the prime to divide n . $\square$

An eight-worker audit checks this theorem over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$ on 9778 tables. It verifies all 1160 nonuniversal quotient classes and a separate implementation reconstructs 48890 modular ranks, with no errors.

We next use congruences in the quasigroup algebra with multiplication and the two division operations. All congruence blocks have equal size and form a Latin quotient.

Theorem 4.5 (Congruence obstructions). *Let L be a finite Latin square.*

- (i) *If L has a congruence with odd block size greater than one, then $\text{pat}(L) = \text{TTT}$.*
- (ii) *If the congruence blocks have size two and the quotient is Q , then $\text{pat}(L) = \text{pat}(Q)$ componentwise.*

Proof. For (i), choose distinct rows x, x' in one congruence block A and any column block B . The two left translations map B bijectively onto the same product block. Thus $L_{x'}^{-1}L_x$ preserves B and is fixed-point-free there. Since $|B|$ is odd, its restriction has a nontrivial odd cycle. Applying the same argument to the two relevant parastrophes proves the column and symbol bits.

For (ii), label every two-element block by $(i, 0), (i, 1)$, where $i \in Q$. The unique form of a binary Latin square gives a function $f: Q^2 \rightarrow \mathbb{F}_2$ such that

$$(i, a) * (j, b) = (i * j, a + b + f(i, j)).$$

Rows in the same quotient block induce only transpositions inside the column fibres. For rows in different quotient blocks, the lift above a quotient cycle of length ℓ is either two cycles of length ℓ or one cycle of length 2ℓ , according to its binary voltage. Thus an ambient odd cycle can occur only above an odd quotient cycle.

Conversely, suppose quotient rows i, i' induce an odd cycle C of length $\ell > 1$. Above a quotient column $j \in C$, the lifted row permutation comparing (i, a) with (i', a') has the form

$$(j, b) \mapsto (\pi(j), b + a + a' + f(i, j) + f(i', \pi(j))).$$

Its voltage around C is $a + a' + S$, because ℓ is odd, where S is independent of a, a' . Choose $a + a' = S$. The voltage is then zero, so the lift of C consists of two cycles of the same odd length ℓ . Hence a quotient row witness lifts to an ambient row witness. This proves equality of the row bits. The same complete argument in the parastrophes proves equality of the column and symbol bits. $\square$

Corollary 4.6 (Order- $2p$ simplicity). *Let p be an odd prime. Every quasigroup of order $2p$ with at least one F view is congruence-simple, and every isotope of it is congruence-simple.*

Proof. A nontrivial congruence has block-size/quotient-order pair $(p, 2)$ or $(2, p)$. The first is excluded by Theorem 4.5(i). In the second, the quotient has pattern TTT by Proposition 3.1, so (ii) excludes it. Both congruence cases force TTT, contradicting even one F view. Finally, the full view pattern is isotopy invariant by Proposition 2.9, and the same argument applies to every isotope. $\square$

Corollary 4.7 (Order- $2p$ line-incidence saturation). *Let $p > 1$ be an odd prime and let L have order $2p$. If even one view of L is F , then M_L has rank $6p - 2$ over every field, all its nonzero Smith invariant factors are 1, and*

$$\mathbb{Z}^{6p} / \text{im}_{\mathbb{Z}} M_L \cong \mathbb{Z}^2.$$

In particular, this holds for every hypothetical order-10 FFF square.

Proof. By Theorem 4.4, a torsion prime must divide $2p$. A characteristic-two defect yields, after isotopy, a congruence with two blocks of odd size p , so Theorem 4.5(i) gives pattern TTT. A characteristic- p defect yields blocks of size two; part (ii) identifies the pattern with that of the odd-order quotient, again TTT by Proposition 3.1. Isotopy invariance contradicts the assumed F view in either case. Thus no Smith prime remains, proving saturation and the displayed cokernel. $\square$

Thus a hypothetical order-10 FFF square must be isotopically congruence-simple. This removes every congruence-imprimitive quasigroup construction, but it does not rule out simple quasigroups and therefore does not decide order 10.

The binary equality iterates through congruence series.

Corollary 4.8 (Prime-factor congruence series). *Suppose that a quasigroup admits a chain of congruence quotients ending in the one-element quasigroup, with every congruence block size prime. It is FFF if all those primes are 2, and it has pattern TTT if at least one is odd.*

Proof. A binary step preserves the entire pattern by Theorem 4.5(ii). At an odd step, Theorem 4.5(i) gives TTT independently of the lower quotient. Iterating proves both statements. $\square$

This characterizes FFF within this congruence-decomposable class. It does not imply the former global power-of-two conjecture, which is refuted by the order-12 witness in Computational Theorem 7.1.

The same theorem also gives a census-free nongroup example.

Theorem 4.9 (Explicit nongroup FFF loop). *On $\mathbb{Z}_4 \times \mathbb{F}_2$, define*

$$(i, a) * (j, b) = (i + j, a + b + \mathbf{1}_{(i,j)=(1,1)}),$$

with coordinates taken modulo 4 and 2, respectively. This is an FFF loop of order 8 that is not group-isotopic.

Proof. The identity is $(0, 0)$, and the first-coordinate fibres form a congruence with quotient C_4 . The quotient is FFF by Corollary 3.3, so binary pattern equality makes the loop FFF. It is not associative because

$$((1, 0) * (1, 0)) * (2, 0) = (0, 1), \quad (1, 0) * ((1, 0) * (2, 0)) = (0, 0).$$

At the identity base row, the basis family in Proposition 3.4 is the family of left translations. Closure would force $L_x L_y = L_{x*y}$ and hence associativity. The displayed failure is therefore a closure failure, so the loop is not group-isotopic. $\square$

5 Direct products

Let L be a Latin square on X and K a Latin square on Y . Their direct product $P = L \times K$ is the Latin square on $X \times Y$ defined by

$$P((x, y), (x', y')) = (L(x, x'), K(y, y')).$$

Lemma 5.1 (Product-cycle formula). *Let π and τ be permutations. If x lies on a π -cycle of length m and y lies on a τ -cycle of length n , then (x, y) lies on a cycle of length $\text{lcm}(m, n)$ under $\pi \times \tau$.*

Proof. The pair returns after $t > 0$ steps exactly when both $m \mid t$ and $n \mid t$. The least such t is $\text{lcm}(m, n)$. $\square$

Theorem 5.2 (Pattern product theorem). *For arbitrary Latin squares L and K ,*

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K)$$

componentwise.

Proof. Rows of P are indexed by (a, α) , and their row maps are $r_{(a, \alpha)}^P = r_a^L \times r_\alpha^K$. Hence the induced permutation between (a, α) and (b, β) is

$$\rho_{(a, \alpha), (b, \beta)}^P = \rho_{a, b}^L \times \rho_{\alpha, \beta}^K.$$

If one coordinate of the two product rows is equal, the corresponding factor is the identity. The identical calculation for column maps gives $\kappa^P = \kappa^L \times \kappa^K$.

For the symbol view,

$$s_{(u, \eta)}^P(c, d) = (s_u^L(c), s_\eta^K(d)),$$

and therefore $\sigma^P = \sigma^L \times \sigma^K$. Thus the same argument applies in all three views.

If a factor has a nontrivial odd cycle in one view, keep the other factor line fixed. The identity contributes cycle length 1, so the product contains the same odd cycle. Conversely, suppose a product induced permutation has an odd cycle of length $\ell > 1$. By Lemma 5.1, $\ell = \text{lcm}(m, n)$ for factor cycle lengths m, n . Both are odd, and at least one is greater than one. That factor already has a nontrivial odd cycle in the same view. This proves the Boolean OR formula in each view. $\square$

Corollary 5.3. *$L \times K$ is FFF if and only if both L and K are FFF. In particular, product with a finite 2-group table preserves the full pattern of the other factor.*

Proof. The first assertion is the all- F case of Theorem 5.2. For the second, combine the same theorem with Corollary 3.3. $\square$

Theorem 5.4 (Group-isotopy product theorem). *$L \times K$ is group-isotopic if and only if both L and K are group-isotopic.*

Proof. At a product base row (b, c) , the basis family factors as

$$H_{L \times K}(b, c) = H_L(b) \times H_K(c).$$

If both factor families are closed, their Cartesian product is closed. Conversely, both factor families contain the identity. If the product family is closed, then $(h_1, \text{id})(h_2, \text{id})$ lies in it for all $h_1, h_2 \in H_L(b)$, proving closure of $H_L(b)$; using (id, k_i) proves closure of $H_K(c)$. Apply Proposition 3.4. $\square$

Corollary 5.5. *Let L be a non-group-isotopic FFF Latin square of order n . For every $m \geq 0$, the product $L \times C_2^m$ is a non-group-isotopic FFF Latin square of order $n2^m$.*

Proof. The table of C_2^m is FFF by Corollary 3.3; apply Corollary 5.3 and Theorem 5.4. $\square$

6 Computational results

The following statements are exhaustive finite computations. Their frozen inputs, programs, commands, outputs, and hashes are specified in Section 15.

Computational Theorem 6.1 (Small orders). *The unique reduced Latin square of order 2 is FFF, and all four reduced Latin squares of order 4 are FFF. None of the 9408 reduced Latin squares of order 6 is FFF. More precisely, their patterns are*

$$6600\text{ TTT}, \quad 936\text{ TFF}, \quad 936\text{ FTF}, \quad 936\text{ FFT}.$$

The exact-cover generator loads no cached square list. Every generated square was checked to be Latin and reduced, all 9408 table strings were distinct, and the generated set agrees exactly with the ANU reduced-order-6 data file [McK]. Proposition 2.9 therefore extends nonexistence from reduced squares to all Latin squares of order 6.

Computational Theorem 6.2 (Order-8 pattern census). *Across the 283657 order-8 main-class representatives, the pattern distribution is*

<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>	<i>Pattern</i>	<i>Count</i>
FFF	230	FFT	81	FTF	40	FTT	315
TFF	133	TFT	209	TTF	232	TTT	282417

Among the 230 FFF main classes, exactly 5 are group-isotopic and 225 are not group-isotopic.

Computational Theorem 6.3 (Subsquare census of the FFF classes). *Among the 230 FFF representatives, the number of intercalates ranges from 2 to 112. Each endpoint is attained by exactly one representative: local source line 10776 (ANU record 10775) is the unique minimum class and is not group-isotopic, while local line 1 (ANU record 0) is the unique maximum class and is group-isotopic. Local physical lines are numbered from one; ANU records are numbered from zero. No representative contains an order-3 subsquare. The distribution of the number of order-4 subsquares is*

$$0^{63}, \quad 4^{156}, \quad 12^{10}, \quad 28^1,$$

where exponents count representatives.

The subsquare validator finds no violation of Theorem 4.1. The multiples of four in the last distribution also follow directly: an order- m subsquare inside an order- $2m$ square forces the three complementary blocks to be Latin subsquares, so half-order subsquares occur in complementary quartets.

The ordered non-symmetric labels refer to the representatives in the cited ANU file. Parastrophy permutes the three bits, so labels such as FFT are not themselves main-class invariants. The FFF count, the existence of every pattern, and the group-isotopy split are unaffected.

The scan tests every unordered pair of lines in all three views. A separate closure-family computation reproduces the 5/225 split using Proposition 3.4. The frozen ANU input is an exact byte match to the file served by the cited URL on August 9, 2026.

Corollary 6.4. *For every $m \geq 3$, Latin squares of order 2^m realize all eight patterns. Every such order also contains a non-group-isotopic FFF Latin square.*

Proof. Computational Theorem 6.2 supplies one order-8 square of each pattern. Take direct products with the FFF group table of C_2^{m-3} and use Theorem 5.2. For the second assertion, start from the explicit loop in Theorem 4.9; non-group-isotopy is preserved by Theorem 5.4. $\square$

This is an existence result only. It neither counts higher-order main classes nor asserts that products of distinct order-8 classes remain distinct.

7 Explicit non-power-of-two witnesses and a dyadic lift bound

The computational nonexistence result at order 10 does not extend to all even non-powers of two. The following finite claims are checked by direct three-view cycle scans, not by an uncompleted solver search.

Computational Theorem 7.1 (Explicit non-power-of-two witnesses). *There are FFF Latin squares of orders 12, 14, and 36. The displayed order-12 table is reduced. Thus the former conjecture that FFF squares of order greater than one exist only at powers of two is false.*

L_{12}	0	1	2	3	4	5	6	7	8	9	10	11
0	0	1	2	3	4	5	6	7	8	9	10	11
1	1	0	3	2	5	4	7	6	10	11	8	9
2	2	3	0	1	7	6	5	4	9	8	11	10
3	3	2	1	0	6	7	4	5	11	10	9	8
4	4	7	5	6	8	11	9	10	3	0	2	1
5	5	6	4	7	9	10	8	11	2	1	3	0
6	6	5	7	4	10	9	11	8	1	2	0	3
7	7	4	6	5	11	8	10	9	0	3	1	2
8	8	9	10	11	3	2	1	0	6	7	4	5
9	9	8	11	10	0	1	2	3	4	5	6	7
10	10	11	8	9	2	3	0	1	5	4	7	6
11	11	10	9	8	1	0	3	2	7	6	5	4

Table 1: An explicit reduced order-12 FFF witness. Every row and column is a permutation; all 198 induced line-pair permutations across the three views have only even cycles.

For odd $m > 1$, let $h(m)$ be the least $a \geq 0$ for which an FFF square of order $m2^a$ exists. The odd-order obstruction, the order-6 computation, and Computational Theorem 7.1 give $h(3) = 2$ and $h(7) = 1$; the order-36 table gives $h(9) \leq 2$. Nothing here decides existence at order 18.

The classical polygon-and-centre round-robin factorisation gives a separate, particularly simple family. Cheng and Sgueglia [CS26, Introduction] recall this factorisation and its prime-order perfectness; the following three-view FFF criterion is proved directly, without claiming priority for the construction or an exhaustive literature classification.

Theorem 7.2 (Prime round-robin family). *Let p be an odd prime and $X = \mathbb{F}_p \cup \{\infty\}$. Define M by $M(\infty, \infty) = \infty$, $M(\infty, x) = M(x, \infty) = x$ for finite x , $M(x, x) = \infty$ for finite x , and $M(x, y) = (x + y)/2$ for distinct finite x, y . Then M is Latin and is FFF if and only if $p \equiv 3 \pmod{8}$.*

Proof. In each finite row a , the affine map $y \mapsto (a + y)/2$ takes the value a only at $y = a$. Replacing that entry with ∞ and placing a at column ∞ makes the row a permutation. The infinity row is the identity. Symmetry gives the Latin property for columns as well.

Write R_a for a row permutation and $d = \text{ord}_p(2)$. R_∞ is the identity, while R_a swaps a and ∞ and acts, after translating a to zero, as multiplication by $1/2$ on $\mathbb{F}_p^*$. Hence an infinity/finite row pair has type $2d^{(p-1)/d}$. For distinct finite a, b , put $P = R_b^{-1}R_a$. Apart from a , $2b - a$, and ∞ , the map is translation $x \mapsto x + a - b$. At the exceptional inputs,

$$P(a) = b, \quad P(2b - a) = \infty, \quad P(\infty) = 2a - b,$$

and $P(b) = a$. In the single p -cycle of the translation, the segment $2b - a \rightarrow b \rightarrow a \rightarrow 2a - b$ is replaced by the 2-cycle (a, b) and the connection $2b - a \rightarrow \infty \rightarrow 2a - b$. The remaining cycle has length $p - 1$, including when $p = 3$. Thus all finite/finite row pairs have type $(2, p - 1)$, and the row view is F exactly when d is even. Symmetry gives the identical column condition.

For a symbol s , let T_s send a column to the row containing s in it. Then T_∞ is the identity; for finite s ,

$$T_s(\infty) = s, \quad T_s(s) = \infty, \quad T_s(x) = 2s - x \quad (x \in \mathbb{F}_p \setminus \{s\}).$$

Thus each T_s is a fixed-point-free involution. An infinity/finite symbol pair has type $2^{(p+1)/2}$. Affine relabelling of the finite points reduces any two distinct finite symbols to $0, 1$. The relative permutation T_1T_0 has precisely the cycles

$$(0, 1, 3, 5, \dots, p - 2), \quad (\infty, 2, 4, 6, \dots, p - 1),$$

both of length $(p + 1)/2$. Hence the symbol view is F exactly when $p \equiv 3 \pmod{4}$.

For $p \equiv 7 \pmod{8}$, the supplement $(2/p) = 1$ and Euler's criterion make d divide the odd number $(p - 1)/2$, so the row and column views are T. For $p \equiv 3 \pmod{8}$, $(2/p) = -1$ makes d even; the symbol view is also F. Primes $p \equiv 1 \pmod{4}$ already fail in the symbol view. This proves the equivalence. $\square$

By Dirichlet's theorem, infinitely many primes satisfy $p \equiv 3 \pmod{8}$; apart from $p = 3$, their orders $p + 1$ are not powers of two. At $p = 19$ the table has order 20. Together with the odd-order obstruction Proposition 3.1 and the separately audited computational exclusion of order 10, this gives the sharp threshold $h(5) = 2$. This last equality depends on the exact order-10 computation, whereas Theorem 7.2 is algebraic.

A second class at order 20. The classical no-hexagon STS(19), with explicit starters in Erskine–Griggs [EG24, Section 3.2] and classification context in Colbourn et al. [CFG⁺10], yields a Steiner loop whose 570 two-line permutations have only even cycles. Its three-view cycle profile has no 18-cycle, whereas the midpoint table at $p = 19$ has type $(2, 18)$ in every row pair. Since the combined cycle-type profile is invariant under paratopy, these two examples are not in the same main class. This initial two-class control does not reproduce the external STS classification. The separate trade-family audit in Section 10 strengthens the represented class lower bound to 514.

Theorem 7.3 (Three-point lift bound). *For every odd integer $m > 1$, there is an FFF Latin square of order $m2^{\text{ord}_m(2)}$. Consequently*

$$h(m) \leq \text{ord}_m(2),$$

and FFF squares exist at every order $m2^a$ with $a \geq \text{ord}_m(2)$.

Proof. The case $m = 3$ is supplied by the order-12 table above, since $\text{ord}_3(2) = 2$. Suppose $m > 3$. Set $e = \text{ord}_m(2)$, take $K = \mathbb{F}_{2^e}$, and choose $r \in K^\times$ of multiplicative order m . Its existence follows from $m \mid (2^e - 1)$ and cyclicity of $K^\times$.

For coefficients $s_i \in K$, $i \in \mathbb{Z}_m$, define a multiplication on $\mathbb{Z}_m \times K$ by

$$(i, a) * (j, b) = (i + j, a + r^i b + s_i r^j). \quad (4)$$

The quotient coordinate and the nonzero coefficient r^i make this a Latin square. Put $t_i = r^{-i} s_i$ and regard t_i as a function $t(r^i)$ on $H = \langle r \rangle$. For a coset C of any nontrivial subgroup of H , write

$$S_C = \sum_{x \in C} x t(x), \quad T_C = \sum_{x \in C} x^{-1} t(x).$$

We first verify that pairwise distinct t_i and nonzero S_C, T_C for all such cosets imply FFF in all three views.

Two distinct lines in the same quotient fiber have relative permutations that are nonzero translations of K : for rows the translation is $r^{-i}(a + a')$, for columns it is $r^i(b + b')$, and for symbols it is $r^{-(k-j)}(u + u')$ in each column fiber. Each is an involution without fixed points. For distinct quotient rows i, k , put $d = i - k$ and let ℓ be the order of d in $\mathbb{Z}_m$. Directly composing their row maps gives

$$R_{(k, a')}^{-1} R_{(i, a)}(j, b) = (j + d, r^d b + r^{-k}(a + a') + r^{j-k}(s_i + r^d s_k)).$$

The quotient shifts by d ; its ℓ -step fiber return is

$$b \mapsto b + r^{j-i}(s_i + r^d s_k) = b + r^j(t_i + t_k).$$

For distinct quotient columns j, l , the quotient shift is $j - l$ and the one-step map, with $d = j - l$, is

$$C_{(l, b')}^{-1} C_{(j, b)}(i, a) = (i + d, a + r^i(b + b') + r^j s_i + r^l s_{i+d}).$$

Its return on the associated coset C is

$$a \mapsto a + (r^j + r^l) S_C.$$

For distinct quotient symbols $(k, u), (l, v)$, use the symbol-line map

$$S_{(k, u)}(j, b) = (k - j, u + r^{k-j} b + s_{k-j} r^j).$$

With $d = l - k$, one step of $S_{(l, v)}^{-1} S_{(k, u)}$ is

$$(j, b) \mapsto (j + d, b + r^{j-k}(u + v) + (1 + r^d) r^j t_{k-j}).$$

Summing over a quotient cycle gives the return

$$b \mapsto b + (1 + r^{l-k}) r^k T_C.$$

In each calculation the line-label terms vanish because a nontrivial quotient cycle has odd length $\ell > 1$ and the relevant geometric sum of an ℓ th root other than one is zero. Every displayed multiplier outside S_C or T_C is nonzero. A nonzero characteristic-two translation has order two, so each lifted cycle has even length 2ℓ . This establishes the asserted three-view criterion.

It remains to choose the coefficients. Put $D = r + 1$, $u = D^{-1}$, and start with the injective function $t_0(1) = 1$ and $t_0(x) = (x + 1)^{-1}$ for $x \in H \setminus \{1\}$. Its values at $1, r, r^2$ are $1, u, u^2$, which are distinct for $m > 3$. Cyclically permute these three values:

$$t(1) = u, \quad t(r) = u^2, \quad t(r^2) = 1, \quad t(x) = (x + 1)^{-1} \quad (x \notin \{1, r, r^2\}),$$

and set $s_i = r^i t(r^i)$. Injectivity is retained.

Let $C = aJ$ be a coset of a subgroup J of odd order $\ell > 1$, and write $c = a^\ell$. If $1 \notin C$, then $\prod_{x \in C} (X + x) = X^\ell + c$ and logarithmic differentiation at $X = 1$ gives $\sum_{x \in C} (1 + x)^{-1} = (1 + c)^{-1}$. Also $\sum_{x \in C} x^{-1} = 0$. Any coset untouched by the patch therefore has

$$S_C = \frac{c}{1+c}, \quad T_C = \frac{1}{1+c},$$

both nonzero.

Now let J be proper, set $z = r^\ell$, and let $A = (\ell - 1)/2 \pmod{2}$, interpreted as an element of K . Its three affected cosets J, rJ, r^2J are distinct. The value changes relative to t_0 are $u + 1$, r/D^2 , and r^2/D^2 , respectively. Pairing x with x^{-1} on $J \setminus \{1\}$ gives $\sum_{x \in J \setminus \{1\}} (1 + x)^{-1} = A$. Including $t_0(1) = 1$, the unpatched moments on J are $S_J(t_0) = A + 1$ and $T_J(t_0) = A$. The other two cosets use the preceding unpatched formula. Adding the appropriate value changes yields

C	S_C	T_C
J	$A + u$	$A + 1 + u$
rJ	$\frac{z + r^2}{(1 + z)D^2}$	$\frac{z + r^2}{(1 + z)D^2}$
r^2J	$\frac{[z(r^2 + r + 1) + r^2]^2}{(1 + z^2)D^2}$	$\frac{z^2 + r^2}{(1 + z^2)D^2}$

All denominators are nonzero. The J entries are nonzero because $u \notin \{0, 1\}$. The rJ numerator could vanish only if $\ell \equiv 2 \pmod{m}$, and the last T numerator only if $\ell \equiv 1 \pmod{m}$; neither is possible for a proper divisor $1 < \ell < m$. The only possible failure is therefore

$$r^\ell(r^2 + r + 1) = r^2. \quad (5)$$

At most one exponent modulo m satisfies this equation. If it holds for a proper divisor ℓ , replace r by r^{-1} . A bad proper divisor j for the inverse would require $r^2 + r + 1 = r^j = r^{2-\ell}$, so $j = m + 2 - \ell > m/2$. Every proper divisor of odd m is at most $m/3$; there is no such j . Thus either r or its inverse makes every proper coset moment nonzero.

For the full-group coset H , all three patched points lie together. With $A = (m - 1)/2 \pmod{2}$, direct summation gives

$$S_H = 1 + A + r^2 + \frac{r}{r+1}, \quad T_H = A + \frac{r}{r+1}.$$

When $m \equiv 3 \pmod{4}$, these are $r(r^2 + r + 1)/(r + 1)$ and $1/(r + 1)$; the first could vanish only if r had order three. When $m \equiv 1 \pmod{4}$, they are $(r^3 + r^2 + 1)/(r + 1)$ and $r/(r + 1)$. The cubic is irreducible over $\mathbb{F}_2$ and its roots have order seven, incompatible with $m \equiv 1 \pmod{4}$. Hence both moments are nonzero. The three-view criterion proves that (4) is FFF.

Finally, multiplying by the table of an elementary abelian 2-group preserves FFF by Corollary 5.3, giving every exponent $a \geq e$. $\square$

Remark 7.4 (The exceptional affine base case). At $m = 3$, over the prescribed minimal field $K = \mathbb{F}_{2^{\text{ord}_3(2)}} = \mathbb{F}_4$, the affine formula (4) cannot be FFF, although Table 1 gives an order-12 FFF square. Put $t_i = r^{-i} s_i$. Row-F requires distinct t_i ; column-F and symbol-F require $S = t_0 + r t_1 + r^2 t_2 \neq 0$ and $T = t_0 + r^2 t_1 + r t_2 \neq 0$. Common translation and nonzero scaling give $(t_0, t_1) = (0, 1)$. Since $r^2 + r + 1 = 0$, $t_2 = r$ forces $T = 0$, whereas $t_2 = r^2$ forces $S = 0$. This argument concerns $\mathbb{F}_4$, not the same formula over all larger binary fields.

As a scope control, over $\mathbb{F}_{16} = \mathbb{F}_2[x]/(x^4 + x + 1)$ take $r = x^2 + x$, $(t_0, t_1, t_2) = (0, 1, x)$, and $s_i = r^i t_i$. Direct three-view checks give an FFF square of order 48. This is not a new order-48 existence claim: products of the order-12 witness already supply that order.

The finite checks in the associated companion run evaluate all subgroup coset moments for $m = 5, 7, 9, 15, 21$ and all line-pair cycles at orders 56, 80, and 240. They control the implementation; the universal quantifier in Theorem 7.3 rests on the argument above. The theorem does not determine $h(m)$ in general, resolve order 18, classify higher-order main classes, or establish literature priority.

8 A common binary fiber over prime-square quotients

The preceding bound may require a larger binary exponent when the odd part is a prime square. The following construction uses a single binary fiber over a two-dimensional odd quotient. It is an internally audited research result awaiting independent mathematical review, not a claim of literature priority.

Theorem 8.1 (Quadratic-norm prime-square lift). *Let p be an odd prime, $e = \text{ord}_p(2)$, and D a positive multiple of e . If $q = 2^D > 3(p^2 - 1)$, then there is an FFF Latin square of order $p^2 q$. Every relative two-line cycle in every view has length 2 or $2p$. In particular, $2^e > 3(p^2 - 1)$ implies $h(p^2) \leq e$.*

Proof. Let $G = \mathbb{F}_p^2$, choose a nonsquare $\nu \in \mathbb{F}_p$, and put $B(x, y) = x_1 y_1 - \nu x_2 y_2$ and $Q(x) = B(x, x)$. The norm Q is anisotropic: if $Q(d) = 0$ and $d_2 \neq 0$, then $\nu = (d_1/d_2)^2$, a contradiction; the case $d_2 = 0$ is immediate. Choose ω of order p in $K = \mathbb{F}_q$ and write $\chi(t) = \omega^t$. For a twist array $c : G \times G \rightarrow K$, define

$$(x, a) * (y, b) = (x + y, \chi(2B(x, y))a + \chi(Q(x))b + c(x, y)). \quad (6)$$

Both fiber coefficients are nonzero, so this is Latin for every c .

We calculate the relative action of two lines with distinct quotient coordinates. For rows x, z , put $d = x - z$ and $y_t = y + td$. For columns y, z , put $d = y - z$ and $x_t = x + td$. For symbols k, l , put $d = l - k$, $y_t = y + td$, and $x_t = k - y_t$. In each case the quotient orbit has length p . Solving (6) and then the inverse line map gives a fiber step

$$w_{t+1} = \chi(D_t)w_t + \chi(E_t)a + \chi(F_t)b + \chi(U_t)c(C_{1,t}) + \chi(V_t)c(C_{2,t}).$$

The slope exponents and the two twist cells are

$$\begin{aligned} D_t^{\text{row}} &= Q(x) - Q(z), & (C_{1,t}, C_{2,t}) &= ((x, y_t), (z, y_t + d)), \\ D_t^{\text{col}} &= 2B(x_t, y) - 2B(x_t + d, z), & (C_{1,t}, C_{2,t}) &= ((x_t, y), (x_t + d, z)), \\ D_t^{\text{sym}} &= 2B(x_t, d), & (C_{1,t}, C_{2,t}) &= ((x_t, y_t), (x_t, y_t + d)). \end{aligned}$$

Every $\chi(U_t)$ and $\chi(V_t)$ is nonzero. For clarity in the symbol case, the line map at symbol (k, a) sends a column (y, w) to the unique row $(k - y, u)$, where

$$u = \chi(Q(k - y) - 2B(k - y, y))w + \chi(-2B(k - y, y))(a + c(k - y, y)).$$

Taking the inverse of the second symbol map yields the third line above.

The sum of the p slope exponents is zero in each view: it is constant for rows and affine in t for columns and symbols, and $\sum_{t \in \mathbb{F}_p} t = 0$. The exponent of a label contribution from step t gains the later slopes $\sum_{s > t} D_s$. Expanding the quadratic form gives the following

two accumulated exponents, where x, y denote the starting quotient row and column in the respective line:

	first line label	second line label
row	$2B(x, y) - Q(x) + tQ(d)$	$2B(z, y + d) - Q(x) - tQ(d)$
col	$Q(x) - 2B(x, y) - tQ(d)$	$Q(x) - 2B(x, z) + (t + 1)Q(d)$
sym	$-Q(x) - tQ(d)$	$-Q(x) - 2B(x, d) + tQ(d)$

Since $Q(d) \neq 0$, summing the character of any one of these affine progressions over $t \in \mathbb{F}_p$ gives zero. Thus both line-label coefficients cancel, the p -step slope is one, and the fiber return is $w \mapsto w + \Lambda_C(c)$ for a linear form in the twist cells.

The two cells at each step lie on different quotient lines. Within a line the p visited cells are distinct, also for symbols because a cell has a unique quotient symbol. Thus each Λ_C contains exactly $2p$ distinct variables with nonzero coefficients. In particular it is not the zero form. A fixed cell belongs to exactly $p^2 - 1$ such forms in each of the three views: one quotient orbit for each other quotient line. Its total incidence is $3(p^2 - 1)$.

Order the p^4 cells. Enforce $\Lambda_C(c) \neq 0$ when the last cell in its support is assigned. Previously assigned cells then forbid precisely one value for this cell. At most $3(p^2 - 1)$ conditions close at any cell, so $q > 3(p^2 - 1)$ leaves an allowed value. Induction makes every return a nonzero translation of K . Consequently different quotient lines have cycles of length $2p$. Two distinct lines in the same quotient fiber induce a nonzero translation on every complementary fiber, giving only 2-cycles. The argument applies to rows, columns, and symbols, proving FFF. $\square$

Corollary 8.2. *For all but $O((\log X)^2)$ primes $p \leq X$, with $e = \text{ord}_p(2)$, one has $h(p^r) \leq \lceil r/2 \rceil e$ for every $r \geq 1$.*

Proof. Theorem 8.1 supplies the prime-square factor; the order- $p2^e$ instance of Theorem 7.3 supplies an odd residual factor. Direct products preserve FFF, giving the stated bound for each r . An exceptional prime satisfies $2^e \leq 3(p^2 - 1) < 4p^2$, hence $e < 2\log_2 p + 2$. If $p \leq X$ and $N = \lfloor 2\log_2 X + 2 \rfloor$, then p divides $\prod_{j=1}^N (2^j - 1)$. This product has fewer than $\sum_{j=1}^N j = N(N + 1)/2$ distinct prime factors. $\square$

At $p = 11$, $e = 10$ and $3(p^2 - 1) = 360 < 1024$, giving an order-123904 construction. A separate standard-library checker validates all 239580 quotient-return conditions of a frozen compact twist certificate; it does not materialize that Latin table. Fresh smaller controls cover $p = 3, q = 64$ and $p = 5, q = 256$. The finite checks support the computation, whereas the universal quantifiers rest on the proof above. No claim is made about minimal exponents, order 18, or literature priority.

9 Affine prime-successor loops

For an odd prime p and $a \in \mathbb{F}_p \setminus \{0, 1\}$, put $b = 1 - a$. Define $Q_p(a)$ on $\mathbb{F}_p \cup \{\infty\}$ by

$$\infty * x = x * \infty = x, \quad x * x = \infty \ (x \in \mathbb{F}_p), \quad x * y = ax + by \ (x \neq y, \ x, y \in \mathbb{F}_p).$$

In each finite row or column, the affine bijection fixes the line label. Replacing that value by ∞ and placing the old label in the new position proves that this is a Latin loop. Write $[u]_p$ for the least positive residue of a nonzero field element.

Theorem 9.1 (Three-view affine criterion). *The loop $Q_p(a)$ is FFF if and only if*

$$\begin{aligned} [a]_p, [a^{-1}]_p, [b^{-1}]_p \text{ are even,} \\ \text{ord}_p(a), \text{ord}_p(b), \text{ord}_p(-b/a) \text{ are even.} \end{aligned} \tag{7}$$

Proof. Use a generic row parameter t , with $c = 1 - t$. A finite row relative to the identity row consists of a two-cycle and $(p - 1)/\text{ord}_p(c)$ cycles of length $\text{ord}_p(c)$. Translations and nonzero scalings of finite labels are automorphisms and are two-transitive on them. It remains to compare rows 0, 1. Their relative map $P = r_1^{-1}r_0$, with $D = -t/c$, satisfies

$$P(\infty) = D, \quad P(0) = 1, \quad P(c^{-1}) = \infty, \quad P(y) = y + D \quad \text{at every other finite } y.$$

Number the translation cycle by jD , $j \in \mathbb{Z}_p$, and put $k = [-t^{-1}]_p$. Then $c^{-1} = kD$ and $1 = (k + 1)D$. The redirected arrows give cycles of lengths $p - k$ and $k + 1$, both even exactly when $[t^{-1}]_p$ is even.

The column view has parameter b . A finite symbol z , viewed as a map from columns to rows, swaps z, ∞ and otherwise sends y to $a^{-1}z - (b/a)y$. Its parameter is a^{-1} and its other coefficient is $-b/a$; the infinity symbol line is the identity. Applying the row calculation to these three parameters proves (7) in both directions. $\square$

Computational Theorem 9.2 (A residue-class boundary control). *The explicitly defined loop $Q_{97}(36)$ is FFF of order 98. Thus $n \equiv 2 \pmod{8}$, $n > 2$, is not a universal FFF obstruction.*

Indeed, $b = 62$, $36^{-1} = 62$, $62^{-1} = 36$, $-62/36 = 36$ in $\mathbb{F}_{97}$, and $36^3 = 62^3 = -1$. The orders and the three least residues in (7) are even. Primality and these identities are exact finite checks. The materialized table also passes all $3\binom{98}{2} = 14259$ line-pair checks. This is not an order-18 construction.

9.1 A uniform large-prime bound

Theorem 9.3 (Eventual prime-successor existence). *For every prime $p \geq 10^{12}$, more than $p/1000$ parameters produce FFF loops $Q_p(a)$.*

The analytic input is Katz's published mixed-character-sum estimate [Kat08, Theorem 1.5]. Its application and the explicit cutoff are checked below, not inferred from finite tables.

Proof. Put $U = \mathbb{F}_p \setminus \{0, 1\}$. For $p \equiv 3 \pmod{4}$, take the quadratic character χ and the sufficient mask

$$W(a) = \frac{1}{4}(1 - \chi(a))(1 - \chi(1 - a)).$$

For $p \equiv 1 \pmod{4}$, take a character of exact order four and use

$$W(a) = \frac{1}{8}(1 - \chi(a)^2)(1 - \chi(1 - a) + \chi(1 - a)^2 - \chi(1 - a)^3).$$

These are zero-one indicators. In the first case $a, b, -b/a$ are nonsquares. In the second, write $p - 1 = 2^s m$ with m odd. The primitive-root exponent of a is odd, that of b is 2 modulo 4, and that of $-b/a$ is odd. All three multiplicative orders are therefore even in either case. The mask's character expansion has coefficient absolute sum one and constant coefficient at least $1/8$.

Let $\psi(x) = \exp(2\pi ix/p)$ and let χ have order $q \in \{2, 4\}$. Except for the all-zero tuple, the sums

$$S(j, k; u, v, w) = \sum_{a \in U} \chi(a)^j \chi(1-a)^k \psi(ua + v/a + w/(1-a)), \quad 0 \leq j, k < q,$$

have absolute value at most $4\sqrt{p}$. For $u \neq 0$, apply the cited theorem with polynomial ua , rational remainder $v/a + w/(1-a)$, and multiplicative polynomial $a^j(1-a)^k$. All finite poles are simple; the pole order at infinity is below one, and every nonpole zero has nontrivial character multiplicity. With t endpoint poles and e other endpoint zeros, the estimate is $(2t+e)\sqrt{p}$. Deleting the remaining $2-t-e$ endpoints changes the sum by at most that number, so the result is at most $4\sqrt{p}$. For $u=0, v \neq 0$, substitute $z=1/a$; for $u=v=0, w \neq 0$, substitute $z=1/(1-a)$. Each is a bijection on U , gives a nonzero linear phase plus at most one simple finite pole, and changes the multiplicative factor to powers of $z, z-1$, up to a constant. The same argument applies. With zero additive phase, the nontrivial multiplicative sums are Jacobi sums or one-character endpoint sums, bounded by $\sqrt{p}$. The all-zero tuple has sum $p-2$.

Let $E(x)$ indicate nonzero even least residues. Geometric summation gives its normalized Fourier coefficients

$$\hat{E}(0) = \frac{p-1}{2p}, \quad |\hat{E}(h)| = \frac{1}{2p|\cos(\pi h/p)|} \quad (h \neq 0).$$

Pairing $h, p-h$ and using $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ bounds the nonzero coefficient sum by $\sum_{r=0}^{(p-3)/2} (2r+1)^{-1} \leq 1 + \frac{1}{2} \log p$. The full absolute sum is at most $L(p) = 3/2 + \frac{1}{2} \log p$. Expanding $W(a)E(a)E(a^{-1})E((1-a)^{-1})$ therefore bounds its parameter count by

$$N_p \geq \frac{(p-2)(p-1)^3}{64p^3} - 4\sqrt{p}L(p)^3. \quad (8)$$

The first term is at least $(p-5)/64 \geq p/65$ for $p \geq 325$. The function $\sqrt{x}/L(x)^3$ increases once $L(x) > 3$. At $x = 10^{12}$, $\log 10 < 2.303$ gives $L(x) < 15.318$ and $260(15.318)^3 < 935000 < 10^6$, exact rational inequalities. It follows that $N_p > (1 - 0.935)p/65 = p/1000$. For example, the finite rational inequality $\sum_{k=0}^9 (2303/1000)^k/k! > 10$ proves the logarithm bound. $\square$

For $p > 3$ these loops are not group-isotopic. Associativity at $(0, 0, 1)$ would force $b^2 = 1$, hence $b = -1$, and at $(1, 0, 0)$ would force $a = -1$, contradicting $a + b = 1$. A group-isotopic loop is a group by the basis-family closure criterion: since its identity row is the identity permutation, closure gives $L_x L_y = L_{xy}$ after evaluating at the identity, hence associativity.

9.2 Exact classification inside the affine family

Theorem 9.4 (Affine-family main classes). *For $p \geq 5$, the loops $Q_p(a), Q_p(a')$ are main-class-equivalent exactly when*

$$a' \in \mathcal{A}(a) = \{a, 1-a, 1/a, 1/(1-a), a/(a-1), (a-1)/a\}.$$

Distinct parameters are never isomorphic; each loop has automorphism group $\text{AGL}(1, p)$.

Proof. Swapping the inputs and solving the relation for either input give the displayed parameters, including all triples involving ∞ . Thus these relations suffice.

The types of row pairs involving infinity and of two finite rows are

$$(2, \text{ord}_p(b)^{(p-1)/\text{ord}_p(b)}), \quad ([a^{-1}]_p, p+1 - [a^{-1}]_p).$$

They coincide only if $\text{ord}_p(b) = p-1$ and $[a^{-1}]_p \in \{2, p-1\}$. The column version exchanges a, b . Outside $E = \{2, 1/2, -1\}$ both identity lines are intrinsic: each is the unique line with all p incident pair types the same, whereas a finite line has both types. An isotopy between such loops must fix the two identity lines; the loop equations make its three maps equal. Its finite-field bijection f satisfies

$$f(ax + by) = a'f(x) + (1 - a')f(y),$$

also at $x = y$. Subtract $f(0)$ and set one argument to zero. Inverting a, b in the resulting identities proves additivity of $h(x) = f(x) - f(0)$. On a prime field h is multiplication by a nonzero scalar; the identity then forces $a = a'$.

For a paratopy first perform its coordinate permutation, replacing a by $t \in \mathcal{A}(a)$. The set E is a single orbit. If exactly one of t, a' is in E , either its special and finite pair types coincide in some view, giving an invariant mismatch, or both identity lines remain intrinsic and the same argument forces equality. If both are in E , they already lie in the same displayed orbit. This proves necessity. For an isomorphism the identity is fixed without exceptions, so the additive argument always gives $a = a'$ and $f(x) = ux + v$. Conversely every such map with $u \neq 0$, fixing ∞ , is an automorphism. $\square$

Let S_p be the full parameter set satisfying (7). The anharmonic orbits have size six, apart from E of size three and the two roots of $a^2 - a + 1 = 0$ when present. Let e_p, h_p indicate whether these exceptional orbits lie in S_p . The exact number of represented FFF main classes is

$$C_p^{\text{aff}} = (|S_p| + 3e_p + 4h_p)/6.$$

Indeed, the three involutions fix only $1/2, 2, -1$, and the order-three elements fix exactly the two polynomial roots. For $p \geq 5$ these exceptional sets are disjoint. Consequently, Theorem 9.3 gives more than $p/6000$ FFF main classes at every prime-successor order $p+1$ with $p \geq 10^{12}$.

Finite controls construct all 253 parameters for the 13 odd primes through 43, compare all views with the criterion, and check parameter transformations and orbit counts. They do not prove the analytic estimate or its uniform cutoff. At $p = 17$ the criterion excludes only the scalar affine family, not all order-18 squares. The classification is within this family, not a census of all Latin squares. Literature priority and external expert review of the analytic application remain open.

10 Row-fibred extensions and represented main classes

The following construction allows the inner operation to depend on an outer row. Related index-two constructions already occur in the loop literature, for example Johnson's dihedral extension [Joh06]. No priority claim for the construction, its FFF consequences, or the bounds below is made here.

10.1 An exact three-view pattern formula

Let $B : Y \times Y \rightarrow Y$ be Latin and, for each $y \in Y$, let $Q_y : X \times X \rightarrow X$ be Latin. Define

$$M((r, y), (c, z)) = (Q_y(r, c), B(y, z)). \quad (9)$$

All sets are finite and nonempty. The fibre squares need not be isotopic.

Theorem 10.1 (Row-fibred pattern formula). *The table M is Latin, and in the three named views*

$$\text{pat}(M) = \text{pat}(B) \text{ OR } \bigvee_{y \in Y} \text{pat}(Q_y).$$

In particular, M is FFF if and only if B and every Q_y are FFF.

Proof. In a fixed row, both coordinate maps in (9) are bijections. In a fixed column, the second output determines y and then the first determines r , proving the Latin property.

For rows $(r, y), (r', y')$ with $y = y'$, the induced permutation fixes the outer column z and restricts on each such layer to the corresponding row-pair permutation of Q_y . With $y \neq y'$, it projects to the row-pair permutation of B on outer columns. For columns $(c, z), (c', z')$ with $z = z'$, the analogous permutation fixes y and restricts to a column-pair permutation of Q_y ; with $z \neq z'$ it projects to the column-pair permutation of B . For symbols $(s, t), (s', t')$ with $t = t'$, the outer column z is fixed, and the outer row is the unique y such that $B(y, z) = t$. The induced map on inner columns is the symbol-pair permutation of Q_y . Every y occurs for one z . With $t \neq t'$, the outer columns instead follow the symbol-pair permutation of B .

Every fibre witness therefore embeds in the same view of M . In every cross-layer case an upstairs cycle has length divisible by its projected cycle length. Thus, if both the quotient and all fibres are F in that view, no odd cycle is possible upstairs. To finish the equality, a quotient odd cycle must be lifted without increasing its length by an even factor. This requires a choice of inner coordinates.

For quotient rows y, y' , fix an inner column c and an inner row r . Choose r' uniquely so that $Q_y(r, c) = Q_{y'}(r', c)$. The induced upstairs row-pair map fixes c , so every quotient cycle lifts with its original length. For quotient columns z, z' , choose upstairs columns with a common inner column c and fix an inner symbol s . At each outer row y on the cycle, choose r_y by $Q_y(r_y, c) = s$. The induced map carries (r_y, y) around precisely the quotient cycle. Finally, for quotient symbols t, t' , take a common inner symbol s and an inner column c . The symbol-pair map keeps c fixed and follows the quotient symbol cycle on z : in the relevant row y , the inner row is determined by $Q_y(r, c) = s$. Hence the quotient odd cycle survives in this view as well. $\square$

For $|X|, |Y| > 1$, the construction cannot produce FFF of order 18: one factor of every such factorization of 18 is odd and greater than one. The order-one square is vacuously FFF and is not an odd-order obstruction.

10.2 Cycle spectra and the order-20 input

For an unordered pair of distinct lines in a view, sort its cycle lengths as a partition λ . Let $\Sigma_L(\lambda)$ count all such pairs, summed over the three views. Isotopy conjugates or inverts pair permutations; parastrophy also permutes the views. Thus Σ_L is a main-class invariant. Sorting each partition before comparison is essential.

Proposition 10.2 (Product spectra and intercalates). *Let $I(L)$ denote the number of intercalates in a Latin square of order n , and let K have order m . Then*

$$I(L \times K) = m^2 I(L) + n^2 I(K) + 4I(L)I(K).$$

If L is FFF, and D duplicates each part of a partition, then

$$\Sigma_{L \times C_2} = 4D_* \Sigma_L + 3n[2^n]. \quad (10)$$

In particular, distinct spectra remain distinct after any number of products with C_2 .

Proof. A two-cycle in a row-pair permutation is one intercalate. If product rows differ only in the L coordinate, the common K row has m choices and each two-cycle has m copies, giving $m^2 I(L)$. The other one-coordinate case gives $n^2 I(K)$. If both coordinates differ, a pair of factor two-cycles has two alignments of the product rows, each producing two two-cycles, giving the last term. Distinct Latin lines have no fixed points, so there are no other ways to produce a two-cycle.

For (10), in each view the n product-line pairs with equal L coordinate have type 2^n . Each unordered distinct L line pair has four product-line pairs. A cycle of length a times a cycle of length b has $\gcd(a, b)$ cycles of length $\text{lcm}(a, b)$. Since every L part is even, both multiplication by the identity on two points and by its transposition duplicate every part. Summing the three views proves the formula. The map D is injective on partitions, so its pushforward is injective. $\square$

Computational Theorem 10.3 (Frozen order-20 family). *One specified cycle-union trade family of an FFF Steiner-loop table of order 20 represents exactly 512 main classes within that family. Two separately specified quadratic tables represent further classes. Consequently at least 514 FFF main classes of order 20 exist.*

The finite evidence enumerates all $2^{10} = 1024$ masks of the ten transpositions in one row pair. Every resulting table is Latin and FFF under two independent physical three-view scanners. Exactly 512 sorted spectra occur, each at complementary masks. Complementing a mask swaps the two rows globally, so it supplies the upper bound of 512 classes as well as the invariant's lower bound. The two quadratic controls have distinct spectra outside that family; their intercalate counts are 190 and 19, whereas the trade-family counts range from 542 to 676. These are exact table comparisons, not a census of all order-20 squares. Equation (10) alone propagates the lower bound 514 to orders $20 \cdot 2^d$.

10.3 Recovering quotient layers

A *binary three-sorted quotient* of Q is a triple of surjective maps a, b, d from its row, column and symbol sets to $\mathbb{F}_2$ satisfying $d(Q(r, c)) = a(r) + b(c)$. The independent labelling of the three sorts is part of the definition. The corresponding partitions, not their chosen bit labels, are intrinsic.

Lemma 10.4 (Binary quotient descent). *Suppose no Q_y in (9) has a binary three-sorted quotient. Every such quotient of M is constant on each inner layer and descends to a binary quotient of B ; conversely, each binary quotient of B pulls back to one of M .*

Proof. First, any three binary maps satisfying $d(Q(r, c)) = a(r) + b(c)$ are either all constant or all surjective. If a is constant, fixing c and varying r shows that d is constant, and then b is constant. The argument with a, b exchanged is the same; if d is constant, varying either input shows both others constant.

Restrict a quotient triple of M to the block of rows at y , columns at z , and symbols at $B(y, z)$. It satisfies this equation on Q_y . Quotient freeness makes all three restrictions constant. Running over y, z makes the full maps constant on every inner layer, and their constants satisfy the quotient equation on B . Surjectivity is preserved by descending or pulling back. $\square$

For $B = E = (\mathbb{F}_2)^d$, every binary quotient has the form

$$a(y) = \chi(y) + a(0), \quad b(z) = \chi(z) + b(0), \quad d(t) = \chi(t) + a(0) + b(0),$$

where $\chi : E \rightarrow \mathbb{F}_2$ is a nonzero linear character. This follows by evaluating the quotient equation at $(y, 0)$ and $(0, z)$ and subtracting the constants. All $2^d - 1$ characters separate points of E . Their common refinement therefore recovers each inner layer intrinsically, in all three sorts.

Theorem 10.5 (Affine-orbit classification of a fixed family). *Fix representatives $Q_1, \dots, Q_m$ of pairwise distinct main classes of order n , none admitting a binary three-sorted quotient. For each function $f : E \rightarrow \{1, \dots, m\}$ define*

$$M_f((r, y), (c, z)) = (Q_{f(y)}(r, c), y + z).$$

Two members M_f, M_g are main-class-equivalent if and only if $g(Ay + u) = f(y)$ for some $A \in \text{GL}(d, 2)$ and $u \in E$.

Proof. By Lemma 10.4, every main-class equivalence preserves the intrinsic layers, up to a permutation of their three sorts, and induces a paratopism of the outer table of E . Its blocks are colored by the main classes $[Q_{f(y)}]$. A nonconstant coloring is constant along an outer row but not along an outer column or symbol line. Thus a paratopism between nonconstant members must carry the source row sort to the target row sort. A constant coloring can match only the same constant coloring, since the input main classes are distinct.

Every paratopism of $y + z = t$ acts affinely on each sort with a common invertible linear part. Indeed, permuting its three coordinates preserves this relation in characteristic two. After undoing that permutation, write $\gamma(y + z) = \alpha(y) + \beta(z)$. Setting $u = \alpha(0)$, $v = \beta(0)$ gives $\alpha(y) = Ay + u$, $\beta(z) = Az + v$ and $\gamma(t) = At + u + v$, with A additive and bijective. The transported block colors now give $g(Ay + u) = f(y)$.

Conversely, this equality makes the three maps

$$(r, y) \mapsto (r, Ay + u), \quad (c, z) \mapsto (c, Az + v), \quad (s, t) \mapsto (s, At + u + v)$$

an explicit isotopism for any $v \in E$. The fixed choice of the representatives is used in this sufficiency argument. $\square$

Corollary 10.6 (Constructed class counts). *Let $c(h)$ be the number of cycles of h on E . The fixed family in Theorem 10.5 represents exactly*

$$T_d(m) = \frac{1}{|\text{AGL}(d, 2)|} \sum_{h \in \text{AGL}(d, 2)} m^{c(h)}, \quad |\text{AGL}(d, 2)| = 2^d \prod_{i=0}^{d-1} (2^d - 2^i) \quad (11)$$

main classes. If the inputs are FFF, these are FFF classes and hence give a lower bound for the total at order $n2^d$. A weaker bound is $\binom{m+2^d-1}{2^d}$, obtained by distinguishing input multisets.

Proof. Burnside's lemma counts colorings fixed by each affine permutation: they are precisely the colorings constant on its cycles, numbering $m^{c(h)}$. Affine orbits refine the input multisets. The FFF assertion follows from Theorem 10.1. $\square$

Lemma 10.7 (A direct certificate for binary quotient freeness). *For an order- n Latin square Q , let A_Q be its $3n$ by n^2 line-cell incidence matrix: the column for (r, c) has ones on row line r , column line c , and symbol line $Q(r, c)$, and zeros elsewhere. Then Q has no binary three-sorted quotient if and only if $\text{rank}_{\mathbb{F}_2} A_Q = 3n - 2$. A nonsingular $(3n - 2)$ -square minor certifies this equality.*

Proof. The two independent line vectors $(\mathbf{1}, \mathbf{1}, \mathbf{0})$ and $(\mathbf{1}, \mathbf{0}, \mathbf{1})$ lie in $\ker(A_Q^T)$, so the rank is at most $3n - 2$. Their span consists exactly of the constant triples $(a, b, d) = (\alpha, \beta, \alpha + \beta)$. The kernel equation for any triple of binary line maps is $a(r) + b(c) + d(Q(r, c)) = 0$ for every cell. As in the proof of Lemma 10.4, any nonconstant solution has all three maps surjective: if one were constant, fixing one input and using the Latin bijections would force the other two to be constant as well. Thus an additional kernel vector is exactly a binary quotient. Rank-nullity proves the equivalence. A nonsingular minor supplies the matching lower bound on the rank. $\square$

Computational Theorem 10.8 (Specialization from order 20). *Direct 58 by 58 binary minors certify quotient freeness for all 1024 trade-mask tables and the two quadratic controls of Computational Theorem 10.3. In particular, its 514 input main classes are binary-quotient-free. Therefore there are at least $T_d(514)$ FFF main classes of order $20 \cdot 2^d$. These earlier certified bounds remain valid. The first four values are:*

Order	Proven lower bound, not a census
40	132355
80	2942384005
160	3625556867870760586
320	73586711420426563009097417622853741216

Proof. The certificates use all 20 row lines, the first 19 column lines, and the first 19 symbol lines, with 58 specified cell columns for each table. A separate binary elimination checks each minor and its binding to the exact table. All 1026 minors are nonsingular. Lemma 10.7 therefore proves quotient freeness for these inputs. Theorem 10.5 and Equation (11) give the construction count; exact affine-group enumeration through dimension four gives the displayed integers. No physical enumeration of that many output squares is asserted. $\square$

This certificate argument concerns the specified input family, not every order-20 FFF square. It removes C157 from these particular class bounds, not from order-10 nonexistence or $h(5) = 2$. The stronger statement that *every* FFF square of order 20 is binary-quotient-free would still use C157: a binary quotient has balanced fibres of size 10 and hence an FFF subsquare of order 10, which C157 excludes.

Two checked order-160 outputs even have the same input multiset and the same sorted cycle spectrum, but their four-point color sets are an affine plane and an affine tetrahedron. Their affine spans have different dimensions, so Theorem 10.5 separates their main classes. This demonstrates a limitation of the spectrum invariant, not its completeness.

The count is exact only within the fixed family. It does not classify all higher-order FFF squares, justify arbitrary direct-product cancellation, or identify all extensions obtained by changing the chosen representatives within their main classes. The binary-quotient-free hypothesis must be checked on the inputs; it cannot be iterated blindly on outputs that already have binary quotients. Computational Theorem 12.3 below supplies a larger independently rank-checked input family and strengthens the lower bound to $T_d(1703)$; the displayed historical $T_d(514)$ values are not claimed to be the strongest current bounds.

11 Block return maps and a common-fibre construction

The quotient-OR identity for the row-fibred construction must not be extended to arbitrary cell-dependent blocks. The distinction is already visible in the order-12 counterexample.

Theorem 11.1 (Cycle lengths in a Latin block substitution). *Let Q be Latin on A , and let F_{ab} be Latin on a common set B for every $(a, b) \in A^2$. The operation*

$$H((a, x), (b, y)) = (Q(a, b), F_{ab}(x, y))$$

is Latin. In each view, same-outer-label line pairs restrict to block line pairs. For different outer labels, a quotient cycle of length d has a bijective inner return map Ω . Each e -cycle of Ω lifts to one cycle of length de . Consequently a view fails exactly when some block fails in that view, or some odd quotient cycle has an odd return-map cycle for a lifted line pair.

Proof. In a fixed row, the desired outer symbol determines the outer column; the block bijection then determines the inner column. The column argument is identical. For rows $(a, x), (a', x')$, equality of outer symbols sends b to b' with $Q(a, b) = Q(a', b')$. The inner step is $F_{a'b'}(x', -)^{-1}F_{ab}(x, -)$. If $a = a'$, then $b = b'$ and this is exactly a row-pair permutation in one block. Otherwise it projects to the quotient row-pair map. Exchanging the two inputs proves the column statement.

For a symbol (s, z) , a column (b, y) determines the unique outer row a with $Q(a, b) = s$; the inner row is obtained by inverting the symbol line of F_{ab} . Equal outer symbols leave b fixed and give a block symbol-pair permutation. Distinct outer symbols follow the quotient symbol-pair map. Every block and every pair of its lines occur as these labels vary, in all three views.

Above a quotient cycle $(b_0, \dots, b_{d-1})$, write the successive inner bijections as $h_0, \dots, h_{d-1}$. A first return to the starting fibre is $\Omega = h_{d-1} \cdots h_0$. An e -cycle of this map closes after exactly e returns and hence de steps; it cannot close before an outer return. This accounts for all cycles and proves the parity claim. $\square$

Both frozen order-12 examples consist of nine FFF blocks of order 4 over a TTT quotient of order 3. Fresh checks verify the return formula on every physical pair in each view. Thus a failing quotient can be masked by even inner returns. This neither contradicts the row-fibred theorem nor the special two-element-kernel equality.

Theorem 11.2 (Odd cubefree common-fibre lift). *Let $N > 1$ be odd and cubefree, $M = \text{rad}(N)$, and $e = \text{ord}_M(2)$. If $D > 0$ is divisible by e and $2^D > 3(N - 1)$, there exists an FFF square of order $N2^D$. Its two-line cycles have length 2 or $2h$, where h is the additive order of a nonzero quotient difference.*

Proof. Write $N = \prod_p p^{r_p}$, with $r_p \in \{1, 2\}$, and put $G = \prod_p \mathbb{F}_p^{r_p}$ and $R = \prod_p \mathbb{F}_p$. Choose $B_p(x, y) = xy$ in rank one and $B_p(x, y) = x_1y_1 - \nu_p x_2y_2$ in rank two, with ν_p a nonsquare. Let $B = (B_p)_p$ and $Q(x) = B(x, x)$. Anisotropy gives $Q_p(d) \neq 0$ exactly when $d_p \neq 0$. Hence $Q(d)$ in the additive group of R and d in G have the same odd order $h(d)$.

The condition on D gives a primitive M th root in $K = \mathbb{F}_{2^D}$. The additive CRT identification of R with $\mathbb{Z}/M\mathbb{Z}$ gives an injective character $\chi : R \rightarrow K^*$. Define, for a twist array $c : G^2 \rightarrow K$,

$$(x, a) * (y, b) = (x + y, \chi(2B(x, y))a + \chi(Q(x))b + c(x, y)).$$

The two fibre coefficients are nonzero, so every twist gives a Latin square. The row, column and symbol relative-step formulas are precisely those in the proof of Theorem 8.1, now evaluated componentwise in R over a quotient orbit of length $h = h(d)$.

Here is why replacing p by composite h is legitimate. Each slope exponent is constant or affine in the step index. Its nonzero coordinates are supported where $d_p \neq 0$; each such odd prime divides h , and $\sum_{t=0}^{h-1} t = h(h-1)/2 = 0$ in that component. Thus the total slope

is one. Composing the later slopes into each of the two label coefficients gives the same displayed exponents as in that proof, of the form $A + tQ(d)$ or $A - tQ(d)$. This follows by expanding the quadratic form in each active component; in an inactive component $d_p = 0$ and the step is constant. The element $\chi(Q(d))$ has exact order h , so

$$\sum_{t=0}^{h-1} \chi(A \pm tQ(d)) = 0.$$

Both line labels therefore cancel in every view. Every inner return is $w \mapsto w + \Lambda_C(c)$.

Choose one orientation and starting point per unordered quotient pair and orbit; conjugate returns have the same nonvanishing condition. The form Λ_C has $2h$ distinct twist cells with nonzero coefficients. For symbols, the two sets of cells have distinct quotient symbols; this ensures distinctness just as different quotient rows or columns do in the other views. A cell lies in one return form for each other quotient line in each view, a total of $3(N-1)$ forms. Assign twist cells in a fixed order and enforce a form to be nonzero when its last cell is assigned. Each closing form forbids one field value. The field-size inequality leaves an allowed choice at every step.

For different quotient labels the return is now a nonzero translation of a characteristic-two field, hence has only two-cycles; physical cycles have length $2h$. For equal quotient labels and distinct fibre labels a, b , the row, column and symbol translations respectively have nonzero constants

$$\chi(2B(x, y) - Q(x))(a + b), \quad \chi(Q(x) - 2B(x, y))(a + b), \quad \chi(-Q(x))(a + b).$$

They give two-cycles. Every physical pair in all three views is covered. $\square$

For example, $N = 11^2 \cdot 41^2 = 203401$ has $\text{ord}_{451}(2) = 20$ and $2^{20} > 3(N-1)$, giving $h(N) \leq 20$. The product of the two separate prime-square constructions gave the upper bound $10 + 20 = 30$. This improves that particular construction, not a known minimum; the order-213281406976 table is not materialized. The compact finite records cover $N = 9, 15, 45, 75$ and direct return-map controls. Their checks do not replace the general proof.

11.1 Limits of the bilinear method

The finite-field zero-count arguments below are elementary special cases of the classical Chevalley–Warning method [Che35, War35]; their needed details are included.

Proposition 11.3 (Scalar quotient boundary). *Let G be a finite odd abelian group and let $\beta : G^2 \rightarrow K^*$ be a symmetric bicharacter over a characteristic-two finite field. If some twist makes*

$$(x, a) * (y, b) = (x + y, \beta(x, y)^2 a + \beta(x, x)b + c(x, y))$$

row- F , every Sylow subgroup of G is elementary abelian of rank at most two. Conversely every such group admits an FFF instance over a sufficiently large field. This is a boundary of this formula only.

Proof. If $d \neq 0$ and $\beta(d, d) = 1$, composition around a row quotient orbit of odd length h has slope one. Its two label coefficients are nonzero scalar multiples of $\sum_{t=0}^{h-1} \beta(d, d)^{\pm t} = 1$. Whatever the twist, one may choose the second row label to cancel the return translation. This creates an odd physical h -cycle. Thus no nonzero d can have $\beta(d, d) = 1$.

An element u of order p^2 violates this condition at $d = pu$. On a rank-three elementary p -subgroup, the exponent of $\beta(d, d)$ is a quadratic form over $\mathbb{F}_p$ (or is identically zero).

Such a form has a nonzero zero: $\sum 1 - Q(d)^{p-1}$ over $\mathbb{F}_p^3$. Each monomial has a variable of exponent below $p - 1$, so its sum is zero modulo p . The number of zeros is divisible by p and includes zero, hence is at least p . Necessity follows; Theorem 11.2 proves sufficiency for the stated groups. $\square$

Proposition 11.4 (Commuting-matrix fibre bound). *Let p be an odd prime, $r \geq 1$ and $D \geq 0$. In the same formula let $G = \mathbb{F}_p^r$, let the fibre be $V = \mathbb{F}_2^D$, and let β be a symmetric bicharacter with commuting image in $\text{GL}(V)$. Row- F requires $D \geq \text{ord}_p(2)\lceil r/2 \rceil$. If $2^{\text{ord}_p(2)} > 3(p^2 - 1)$, this dimension is attained by an FFF construction within this ansatz.*

Proof. Write $e = \text{ord}_p(2)$. The row-return computation uses only commutativity: the label coefficients are invertible matrices multiplied by $S(q) = \sum_{t=0}^{p-1} q^t$ and $S(q^{-1})$, where $q = \beta(d, d)$. If $q = I$, both sums are I , and a choice of the second row label cancels the return. Thus $\beta(d, d) \neq I$ for every nonzero d is necessary.

The commuting matrices of exponent p diagonalize simultaneously over $\mathbb{F}_{2^e}$. Each nontrivial eigencharacter has a Frobenius orbit of length e , so there are at most $t = \lfloor D/e \rfloor$ such orbits. One representative from each yields a quadratic exponent form Q_i on $\mathbb{F}_p^r$. If $r > 2t$, summing $\prod_i (1 - Q_i(d)^{p-1})$ shows, by the same degree argument as above, that their common zero count is divisible by p . A nonzero common zero would give $\beta(d, d) = I$, a contradiction. Hence $r \leq 2t$. For attainment, split G into rank-two summands and at most one rank-one summand. Apply Theorem 11.2 to each over $\mathbb{F}_{2^e}$ and take their direct product. The resulting block diagonal bicharacter has exactly the required fibre dimension. $\square$

These propositions concern the exact scalar or commuting-matrix formula, not nonlinear blocks, arbitrary quotient operations, or the unrestricted minimum $h(m)$. They provide no new exclusion of order 18. In particular, having a fixed vector is not the same as being the identity matrix and cannot replace the condition used in the last proof.

12 A pattern-preserving affine corner trade

Latin squares with a unique intercalate are an established subject; see Wanless [Wan01]. The result here concerns the three-view FFF pattern of a particular affine trade. Neither the existence of unique-intercalate squares nor priority for this formula is claimed to be new.

For an odd prime p and $a \in \mathbb{F}_p \setminus \{0, -1\}$, define P_a on $\mathbb{F}_p \cup \{\infty\}$ by

$$P_a(r, c) = \begin{cases} \infty & c = ar, \\ r + c & c \neq ar \end{cases} \quad (r, c \in \mathbb{F}_p),$$

with $P_a(r, \infty) = (a + 1)r$, $P_a(\infty, c) = (a + 1)c/a$, and $P_a(\infty, \infty) = \infty$. This is the standard one-point prolongation of addition along its affine transversal. Let P'_a swap the two entries in each of rows $0, \infty$ at columns $0, \infty$. Those four cells form an intercalate.

Theorem 12.1 (Corner trade). *Both tables are Latin and $\text{pat}(P'_a) = \text{pat}(P_a)$ component-wise. If $p \geq 5$ and $a \notin \{1, -2, -1/2\}$, then $I(P_a) = p$ and $I(P'_a) = 1$.*

Proof. The prolongation is Latin, and an intercalate swap preserves this property. Put $b = (a + 1)/a$, $d = \text{ord}_p(b)$ and let a_0 be the integer representative of a in $\{1, \dots, p - 2\}$. The autotopisms

$$(r, c, s) \mapsto (r + t, c + at, s + (a + 1)t)$$

fix infinity. Together with the scaling autotopisms $(r, c, s) \mapsto (ur, uc, us)$ for $u \neq 0$, they make all distinct finite-finite row pairs have the same cycle type. For rows $(0, t)$ with $t \neq 0$, scaling columns by t^{-1} gives the relative map

$$0 \mapsto a, \quad \infty \mapsto -1, \quad a+1 \mapsto \infty, \quad x \mapsto x-1 \quad (x \neq 0, a+1).$$

Its cycles containing 0 and infinity are separate, of lengths $a_0 + 1$ and $p - a_0$. For rows (∞, t) , the ordinary finite map is $x \mapsto bx - t$. Prolongation replaces its fixed point at by (at, ∞) ; the remaining cycles have length d . If $t \neq 0$, the point 0 belongs to one of those d -cycles.

Writing $\tau = (0, \infty)$ on columns, the trade precomposes rows $0, \infty$ with τ . The pair $(0, \infty)$ is conjugated and keeps its type. For $(0, t)$ the transposition joins the two cycles into one $(p+1)$ -cycle. For (∞, t) it joins the two-cycle containing infinity to a separate d -cycle, giving lengths $d+2, d, \dots, d$. Pairs of distinct nonzero finite rows are unchanged. Thus row-F holds after the trade exactly when $a_0 + 1, p - a_0, d$ are all even, which is also its condition before the trade. Necessity uses an unchanged nonzero finite pair and the pair $(0, \infty)$; these exist also for $p = 3$.

Transposition gives exactly $P_a^T = P_{a-1}$, preserving the corner trade. In the symbol view, take symbols as lines and columns as positions, and relabel finite columns by $c' = -c$. Solving the table equation gives precisely the same construction with parameter $-a/(a+1)$, again taking the corner to the corner. Applying the row calculation to these two parameters proves column and symbol equality.

Each two-cycle in a row-pair permutation is one intercalate. In P_a the p pairs involving infinity each contribute their distinguished two-cycle. Additional two-cycles occur exactly when $d = 2$, $a_0 + 1 = 2$ or $p - a_0 = 2$, that is, $a = -1/2, 1, -2$. Outside these exceptions the count is p . After the trade, the complete pair list above leaves only the distinguished two-cycle of $(0, \infty)$, giving one. $\square$

Corollary 12.2 (One intercalate and no standard two-point contraction). *For every prime $p \geq 10^{12}$ there exists a non-group-isotopic FFF square of order $p+1$ with exactly one intercalate. It is not isotopic or parastrophic to a standard two-transversal prolongation of an order- $(p-1)$ square with all retained cells unchanged.*

Proof. The isotopy $x = (a+1)r$, $y = (a+1)c/a$, with symbols unchanged, takes P_a to the loop of Section 9 with parameter $\alpha = 1/(a+1)$. The inverse correspondence $a = (1-\alpha)/\alpha$ is bijective. Theorem 9.3 gives more than $p/1000 > 3$ FFF parameters; at most three are exceptional for Theorem 12.1. This proves existence. A group table of even order n with t nonidentity involutions has $tn^2/4$ intercalates: each involution gives $n/2$ unordered row pairs with $n/2$ two-cycles. Here $t \geq 1$ and $n \geq 4$, so this is greater than one. Intercalate count is isotopy invariant.

The unique intercalate of a generic P'_a has rows, columns and symbols $\{0, \infty\}$. Delete them. For each $r \neq 0$, the retained position $(r, -r)$ is a hole because its symbol was zero. Its possible values must lie in

$$\{r, (a+1)r\} \cap \{-r, -(a+1)r/a\}.$$

The four possible equalities respectively force characteristic two, $a = -1/2$, $a = -2$ or $a = -1$, all excluded. Thus the intersection is empty and even one such hole forbids a Latin completion with retained entries fixed. A standard two-transversal prolongation has an order-two corner whose reverse operation is just such a completion. Uniqueness of the intercalate and invariance under all three-sort relabellings and coordinate permutations prove the final assertion. $\square$

This corollary inherits the published character-sum estimate used in Theorem 9.3; it is not a physical computation at enormous orders. Independently scanned small examples are $(p, a) = (13, 3)$ and $(19, 7)$, of orders 14 and 20. The finite checker tests pattern preservation at all 253 admissible parameters over primes through 43 and the intercalate and empty-domain formulas at all 216 generic parameters. Every one of the 15 affine cases at $p = 17$ remains non-FFF. No arbitrary order-18 claim follows.

Computational Theorem 12.3 (Expanded explicit class lower bounds). *There are at least 1703 FFF main classes at order 20, at least $T_d(1703)$ at order $20 \cdot 2^d$, and in particular at least 1450956 at order 40.*

Proof. The frozen enlargement of the Steiner input tests every trade mask for each of its 22 all-two-cycle row pairs. Its historical enumeration records 22528 masks, 19480 FFF labelled mask instances and 1700 distinct exact aggregate spectra. Two quadratic controls add two further spectra. For the present lower bound it suffices to reconstruct one specified table per spectrum: the fresh companion checker reconstructs all 1702, validates Latinness and all three views by two methods, and compares the spectra as exact tuples, not merely hashes. Each has at least 19 intercalates; the checked P'_7 at $p = 19$ has one and supplies another class.

For all 1703 selected tables, fresh binary elimination produces a 58-square minor, and a separate explicit matrix elimination checks its nonsingularity and binding to the table. Lemma 10.7 establishes binary quotient freeness without C157. Apply Theorem 10.5 and Equation (11); at $d = 1$ the count is $1703 \cdot 1704/2 = 1450956$. $\square$

The fresh representative audit proves these lower bounds without repeating the complete 22528-mask search. It does not prove an exact number of all FFF classes, completeness of the spectrum invariant, or literature priority. The older 514-input bounds remain valid but are no longer the strongest bound in this edition.

13 Current search boundaries and research priorities

Order 18 remains undecided. A fixed input, permitted transformation, and complete finite search space must accompany every local negative result. The following developments are not a cumulative percentage of an unrestricted search.

Proposition 13.1 (Intercalates forced by paired blocks). *If a normalized line family in any view of an order- $2m$ Latin square preserves a partition into m pairs, then its intercalate count is $I = m^2 + 2K$ for some nonnegative integer K .*

Proof. Normalize a row view by its base row, identifying columns and symbols. Every row has the form $L(g, (x, t)) = (\phi_g(x), t + f_g(x))$ on $[m] \times \mathbb{F}_2$. Let J flip every symbol pair. The two column maps of block x satisfy $C_{x,1} = JC_{x,0}$. Thus their relative permutation is conjugate to J and has m transpositions. The m within-block column pairs contribute exactly m^2 intercalates. Between two distinct blocks, the relative maps for column bits $(0,0), (1,1)$ are identical, as are those for $(0,1), (1,0)$. Their two-cycle contributions are therefore even. Parastrophy gives the other views without changing the intercalate count. $\square$

At order 18 a paired-block view therefore requires at least 81 intercalates and an odd total. This is a necessary condition, not a proof that a hypothetical FFF18 square cannot have such a view.

Local distance certificates. One frozen FFT18 source has a certified same-label lower bound of 35 changed cells for even a symbol-F Latin target. A different frozen FTF18 source has a bound of 37 cells for even a column-F target. The supporting reductions use physical odd-cycle supports and Latin replacement constraints; their retained UNSAT records have independently checked DRAT traces. This paper update does not rerun those historical proof checkers. Neither bound constructs a target at the stated distance or excludes a more distant target. A small number of failing pairs is not evidence of a small cell-edit distance to FFF.

Distinct normalizations with different coverage proofs. The 14-case longest-zero-cycle normalization in the full report is one complete unrestricted split. A later dual-sign argument gives an alternative 33-case split with positive row-1 and column-1 signs. These are alternatives: the extra column-sign constraint must not be added to the 14 cases without another coverage proof. A further three-positive-sign shortcut is false even on a frozen FFF8 control. The paired baseline/overlay order-18 tests time out; they establish neither a case exclusion nor a runtime improvement.

Other finite-family exclusions. The fixed-remainder contraction audit now covers the entire frozen 22-row-pair trade-mask family and the two quadratic controls, with no FFF18 output. A separate theorem excludes FFF completions of principal fixed-remainder two-point contractions of Steiner loops of order $n \geq 8$ divisible by four: their Latin completions are symmetric of order $n - 2 \equiv 2 \pmod{4}$ and greater than two. Neither statement covers arbitrary parents, nonprincipal Steiner contractions, or repairs changing retained cells. The new unique-intercalate family has its different, direct noncontraction obstruction in Corollary 12.2.

Free-order-three incidence conditions and affine quotient no-goods constrain specified symmetry subclasses only. The recent Mendelsohn, four-transversal and repair pilots do not decide them. These model derivations can be useful without being a breakthrough on unrestricted existence. A shorter equivalent CNF is not, by itself, evidence of a faster solver.

How further searches will be evaluated. Before extending a computation, specify its mathematical hypothesis, covered space, baseline, resource limit and decisive outcome. Separate general theorems, complete finite exclusions and heuristic observations. Use positive and negative physical controls, compare genuinely different methods, and require evidence of benefit before expanding a sequence of similar timeout pilots. Literature comparison and external expert review remain independent tasks. CPU utilization, file counts and claim identifiers are not measures of distance to an order-18 solution.

14 An exact position–value coloring formulation

This section gives an equivalent finite constraint model, not a decision of any new order. In one view let each line a be a bijection $f_a : P \rightarrow Q$. For distinct a, b the permutation $\pi = f_b^{-1} f_a$ is fixed-point-free. A permutation has only even cycles if and only if its points admit a binary coloring that toggles along every edge: alternating a coloring around a cycle is consistent exactly when its length is even.

Theorem 14.1 (Position–value channel). *Introduce colors $e_i, h_v \in \{0, 1\}$ for positions $i \in P$ and values $v \in Q$. The equations*

$$e_i = h_{f_a(i)}, \quad e_i = 1 - h_{f_b(i)} \quad (i \in P) \quad (12)$$

are satisfiable if and only if π has only even cycles. Every solution has exactly $n/2$ ones in each color family. One may add $e_{i_0} = 0$ at any chosen position without changing satisfiability.

Proof. For $j = \pi(i)$, equality $f_a(i) = f_b(j)$ makes (12) imply $e_j = 1 - e_i$. Conversely, from a toggling coloring set $h_v = e_{f_a^{-1}(v)}$. The first equation is immediate. The second follows because $f_a^{-1}f_b = \pi^{-1}$ and toggling also holds on inverse edges. Since the lines are bijections, summing the two equations gives $\sum_i e_i = \sum_v h_v = n - \sum_v h_v = n/2$. Complementing all e and h simultaneously preserves the equations and permits the single gauge condition. $\square$

For a Latin cell variable X_{rcs} asserting $L(r, c) = s$, the antecedents A, B for $f_a(i) = v$ and $f_b(i) = v$ are:

View	Position	Value	A	B
row	column i	symbol v	X_{aiv}	X_{biv}
column	row i	symbol v	X_{iav}	X_{ibv}
symbol	column i	row v	X_{via}	X_{vib}

Writing $e = e_i$ and $h = h_v$, four clauses implement the equations:

$$(\neg A \vee \neg e \vee h), \quad (\neg A \vee e \vee \neg h), \quad (\neg B \vee e \vee h), \quad (\neg B \vee \neg e \vee \neg h).$$

With Latin exactly-one constraints, imposing these clauses independently for every line pair in every view is equivalent to FFF. The reverse implication extends each FFF table by its pairwise alternating colorings; no coupling of arbitrary gauges is required. Isotopy invariance permits reduced tables in an existence search.

Each line pair uses $2n$ colors, $4n^2$ conditional clauses, and one optional gauge unit. Balance follows from the equations and requires no cardinality auxiliaries. Frozen positive controls at orders 8, 12, and 14 and checked negative order-6 controls validate the implementation. In one fixed order-18 row context, the recorded model has 22356 variables and 745065 clauses, compared with 270216 and 3190617 in the prior physical-arc representation. Both bounded 60-second solver probes timed out. This is a size reduction, not evidence of faster solution or of nonexistence at order 18.

15 Computational reproducibility

The research repository separates proofs, frozen data, programs, commands, logs, and SHA-256 manifests. A companion reproducibility quickstart supplies the complete command-to-output map. The compact paper uses the following packages.

This edition’s additional evidence

The dated source package `manuscript/candidates/2026-09-28_research_review/` contains build commands, bounded auditors, selected frozen data and proof notes, and a theorem-to-evidence map. Its intake stops at C271. The public copy is a curated selection;

its own manifests identify the bytes actually shipped. Earlier editions and their evidence are retained as history, not silently resealed.

The preceding review’s 1026 direct binary-rank certificates and larger-field scope control remain included. Its auditor reconstructs those fixed tables, checks all eight order-8 patterns and the stated order-12, 14, 36, 48, 98, and 160 controls. This update separately reconstructs 1703 order-20 representatives, checks each table by independent permutation and matching-component methods, compares exact spectra, and constructs and independently verifies a 58-square binary minor for every table. The new class-growth bound therefore does not inherit C157. Table hashes and minor cell indices are retained, and malformed certificates are rejected by regression tests.

The update also checks every physical line pair of the two order-12 block-substitution examples and all 253 affine corner-trade parameters over primes through 43. Its 216 generic cases test both the intercalate formula and the empty contraction-hole domains. Compact CRT-return checks use separate physical probes for the two line-label coefficients; they do not materialize the enormous example table.

The source-specific order-18 distance certificates, larger trade-mask enumerations and earlier solver results are retained with their original scope. A fresh small representative audit is not a rerun of every historical enumeration. Neither a solver timeout nor a hash agreement is a proof of a larger nonexistence statement.

`ORDER10_REPRODUCIBILITY_GATE.md` records the unresolved durable external route for the full C157 computation. The paper update does not rerun its master searches or establish public reconstruction of their large graph payloads. This remains a separate reproducibility gate.

The two PDF variants have the same evidence boundary and overlapping content; they are not independent studies. Internal AI proof review, independent implementations and machine proof checking are recorded separately from external human refereeing. SHA-256 certifies bytes, not correctness, completeness, novelty, author responsibility or a rights grant. Literature priority and expert review remain open.

Trade parity `repro_runs/2026-08-10_paper_symbol_sign_audit/`. The exact control checks 1028307 row-, column-, and symbol-view trades on all reduced Latin squares of orders 2, 4, and 6, with zero failures. The proof, not this computation, establishes the general theorem.

Small orders `repro_runs/2026-04-26_small_order_witness_reruns/`. The exact-cover generator freshly enumerates 1, 4, and 9408 reduced squares of orders 2, 4, and 6. Every table is checked Latin, reduced, and unique. The generated order-6 set agrees exactly with the official ANU file.

Order-8 census `repro_runs/2026-04-26_order8_core/`. The frozen 5140393-byte input has SHA-256 `e5980ff5bc4ac46cbbefd1f1333258d0905724c42a976c4afe61ea501a4ccf87`. The theorem-relevant output `results/order8_recheck_fullscan_results.json` has SHA-256 `10f2f43c8dcaa8ec93f83abf6b768efa47f909a9c8623d4bc648e6da097ccd39`.

Group-isotopy closure `repro_runs/2026-04-26_order8_closure_reconstruction/`. This independently reconstructs the basis-family closure scan and reproduces the 5/225 split.

Direct products `repro_runs/2026-04-26_direct_product_fff/`. The validator checks the product-permutation formula in all three views and tests the group-isotopy equivalence on group and non-group examples.

Explicit non-power-of-two witnesses `repro_runs/2026-09-23_fff_nonpower_witness_intake/`. An independent standard-library scanner checks the frozen order-12, 14, and 36 tables as Latin, traverses every induced cycle in all three views, and records exact hashes. The displayed reduced order-12 table is `data/order12_first.json`, SHA-256 `8b71d6ecaa6ab86a806a0660529b404d8f3667db2b58b82bf5c72fed010fa416`.

Three-point lift controls `repro_runs/2026-09-23_fff_three_point_lift_audit/`. The script checks subgroup-coset moments for five odd parameters and scans three views at orders 56, 80, and 240. The universal bound is proved in Theorem 7.3; finite controls do not replace it.

Subsquare and congruence obstructions `repro_runs/2026-08-09_fff_subsquare_congruence_obstructions/`. The package scans all subsquares of the 230 frozen FFF representatives, exhausts the binary twists used as controls, and validates the explicit twisted order-8 loop.

Binary line kernel and half-order subsquares `repro_runs/2026-08-11_fff_binary_homotopy_subsquares/`. The package compares the subsquares constructed from every nonuniversal binary line dependency with an independent direct enumeration on 9778 tables. All predicted and observed sets agree exactly, and the semantic verifier reproduces the frozen order-8 distribution with zero errors.

Prime-field line quotients `repro_runs/2026-08-11_fff_prime_field_line_quotients/`. The audit checks five prime fields on 9778 tables, verifies all 1160 nonuniversal quotient classes, and independently reconstructs 48890 modular ranks with zero errors. The theorem itself is proved separately.

External-input release audit `repro_runs/2026-08-09_paper_release_audit/`. It downloads the cited ANU and Wanless inputs afresh and verifies exact byte matches to the frozen files. As with any computational theorem based on an external census, completeness of the published enumeration remains a cited scientific dependency.

Every package contains a README, exact commands, environment record, input and output manifests, and a package-level SHA-256 manifest. The command

```
python3 repro_runs/2026-08-10_paper_symbol_sign_audit/scripts/audit_manuscript_computational_claims.py
```

cross-checks the historical September 16 long manuscript. The compact draft retains its relevant computational statements and adds explicit witness and constructed-class statements. These require the additional dated evidence and checks described above.

Generated PDFs, solver binaries, virtual environments, and raw chat exports are not part of the tracked source archive. No timeout, partial solver model, or partial order-10 diagnostic is used as proof evidence in this compact paper. The complete C157 master computation is a different, explicitly imported dependency.

16 Discussion and open problems

The former assertion that FFF squares of order $n > 1$ exist exactly at powers of two is false: the displayed order-12 witness is a counterexample. Nontrivial odd orders, order 6, and order 10 are excluded; the order-10 argument is an explicit computational dependency

imported from the full companion report. Order 18 remains the next undecided small even non-power-of-two order.

The row-fibred theorem preserves the exact three-view pattern even with different fibres. Under the additional binary-quotient-free hypothesis, Theorem 10.5 classifies the fixed elementary-abelian-quotient family by affine orbits. Computational Theorem 12.3 gives lower bounds for all FFF classes, not exact censuses or unrestricted direct-product cancellation.

The next mathematical questions are:

1. Decide unrestricted FFF existence at order 18 and sharpen the minimal dyadic-exponent function $h(m)$.
2. Find structural obstructions coupling all three views that do not also exclude the explicit order-12, 14, and 98 controls.
3. Determine what lies outside the fixed constructive families and whether quotient recovery survives weaker hypotheses.

The full report records several exact, but restricted, order-18 construction exclusions. None covers all order-18 Latin squares. Timeouts in the smaller equivalent coloring model are likewise not exclusions. No global Alon–Tarsi conclusion is claimed [AT92]; literature priority and independent expert review remain open.

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